

International Conference on
TRANSFORMING
HIGHER EDUCATION FOR THE FUTURE
Education 4.0

Book of Abstracts

THEC 2025



SEEKING EXCELLENCE TOGETHER

This image may appear simple at first glance, but it carries a profound meaning. Each fingerprint represents a fundamental value—diversity of thought, collaboration, and interconnectedness. As these unique patterns intertwine, they create a luminous star, symbolizing excellence achieved through unity and shared purpose.

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INDEX

S. No.	Tracks	Page Number
1.	Student Engagement, Employability and Career Readiness	1
2.	Project-Based Learning	13
3.	Global Implications of Education	25
4.	Faculty Development and Support	31
5.	Experiential Learning	45
6.	Changing Paradigm in Higher Education Scenario	65
7.	Artificial Intelligence (AI) in Education	79
8.	Advances in Educational Technology	95
9.	Active Learning	109
10.	Academia-Industry Connect	135

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Message from the Chief Guest



Fellow, Indian and International Academies

It is an honour and a pleasure to be associated with this important conference on “Transforming Higher Education: Education 4.0.” As we collectively navigate the currents of rapid technological advancement, shifting societal expectations, and global challenges, the time is ripe for a transformative rethinking of how we teach, learn, and innovate.

Education 4.0 is not merely about the integration of digital tools. It represents a bold commitment to democratizing knowledge, decentralizing innovation, and nurturing problem-solvers who are empathetic, entrepreneurial, and ethically grounded. The role of academia is no longer confined to dissemination—it must now pivot toward co-creation, collaboration, and community relevance.

My own journey—spanning fundamental research in microfluidics to the development of frugal diagnostic devices aimed at underserved populations—has taught me that deep science and societal impact must go hand in hand. Today’s students must not only learn concepts but must be empowered to question, create, and catalyze change. This conference offers a valuable platform for such dialogues and directions.

I am heartened to see contributions in areas such as AI-driven learning, sustainable campus initiatives, personalized education pathways, and industry-academia collaboration. These are not just trends—they are imperatives for the future of education.

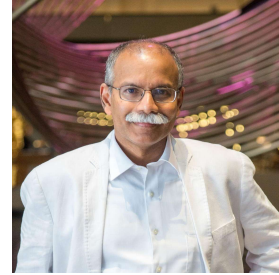
I commend the organizers, contributors, and participants for their vision and effort. May this conference ignite ideas that will reshape educational ecosystems, rejuvenate institutional missions, and rekindle our collective commitment to inclusive and transformative learning.

Warm wishes for a thought-provoking and impactful event.

Prof. Suman Chakraborty

Director, Indian Institute of Technology Kharagpur
Shanti Swarup Bhatnagar Awardee

Message from the Vice-Chancellor



Dear Colleagues, Researchers, and Education Innovators,

It is with great pleasure that I welcome you to this landmark conference on "Transforming Higher Education: Education 4.0" and present this comprehensive collection of abstracts representing cutting-edge research and innovative practices from institutions worldwide.

We stand at a pivotal moment in the evolution of higher education. The Fourth Industrial Revolution has fundamentally reshaped how we live, work, and learn, demanding a corresponding transformation in our educational paradigms. Education 4.0 represents more than technological integration—it embodies a holistic reimagining of pedagogy, curriculum design, student engagement, and institutional governance.

The abstracts compiled in this volume reflect the collective wisdom and forward-thinking approaches of educators, researchers, and practitioners who are actively shaping the future of higher education. From artificial intelligence and personalized learning pathways to sustainable campus initiatives and global collaborative frameworks, these contributions demonstrate how our institutions are evolving to meet the complex challenges of the 21st century.

As we navigate this transformation, we must remain committed to the core values that define quality higher education: academic excellence, critical thinking, innovation, inclusivity, and social responsibility. Education 4.0 is not about replacing these foundational principles but rather enhancing our capacity to deliver them more effectively and equitably to diverse learning communities.

I encourage all participants to engage deeply with these research findings, to challenge conventional wisdom, and to forge new partnerships that will accelerate positive change across our sector. The future of higher education depends on our collective ability to embrace innovation while preserving the intellectual rigor and humanistic values that make our institutions vital centers of learning and discovery.

Thank you to all contributors, organizers, and participants for your dedication to advancing higher education. May this conference serve as a catalyst for meaningful transformation that benefits students, institutions, and society at large.

Prof. Padmakumar Nair

Vice-Chancellor

Thapar Institute of Engineering and Technology, Patiala, Punjab

Message from the Director, NITTTR, Chandigarh



It gives me great pleasure to convey my warm greetings to all the participants, contributors, and readers of the proceedings of the International Conference on “Transforming Higher Education for Future: Education 4.0” (Sept. 26–27, 2025). Higher education worldwide is undergoing a paradigm shift driven by rapid technological advances, changing learner aspirations, and the unique requirements of Industry 4.0. Thus, the Education 4.0 calls for a system that is flexible, digital, inclusive, and outcome-driven, preparing graduates for skills and roles that are constantly evolving.

As one of India’s premier institutions dedicated to the professional development of technical teachers, curriculum innovation, and educational research, NITTTR Chandigarh has a unique role in enabling this transformation. Through our training programs, research initiatives, consultancy services, and digital resources development, we are equipping higher educational institutions to smoothly adopt emerging pedagogies, effectively integrate technology, and align with national priorities such as the National Education Policy (NEP) 2020 and the Sustainable Development Goals (SDGs).

These proceedings represent a valuable compendium of ideas, case studies, and innovations discussed during the conference. They offer actionable insights for educators, policy makers, and institutions striving to reimagine higher education in the era of Education 4.0. I extend my sincere appreciation to the organising team, paper contributors, keynote speakers, and participants for their efforts in making this conference a success. I believe that the papers in the proceedings and deliberations during the conference will catalyse building a future-ready, globally competitive, and socially responsible higher education system.

With best wishes,

Prof. Bhola Ram Gurjar

Director

NITTTR Chandigarh

Message from the Group HR Head, Avantha Group



As the Group HR Head at Avantha Group, I am delighted to be part of this significant milestone in our Institution's continued journey toward educational transformation.

The concept of Education 4.0 resonates deeply with our Board's commitment to foster an environment where innovation fuels academic excellence. The abstracts in this publication reflect the extraordinary caliber of research and thought leadership within the academic community. From artificial intelligence applications in personalized learning to sustainable campus initiatives, each contribution adds valuable insight to the global conversation on educational advancement.

What particularly impresses me is the interdisciplinary nature of the work presented here. The convergence of technology, pedagogy, industry collaboration, and social responsibility demonstrates a holistic approach to educational transformation. These are not isolated academic exercises but practical solutions that will impact real students in real classrooms.

The role of the Executive Council is to facilitate the translation of visionary ideas into sustainable institutional practices. The research presented at this Conference provides the evidence base and innovative framework necessary for informed decision-making and strategic planning.

I congratulate all contributors for their scholarly rigor and creative vision. Your work is instrumental in positioning Indian Institutions as leaders in the Education 4.0 movement.

Mr. Sanjay Sen

Group HR Head, Avantha Group
Thapar Institute of Engineering and Technology, Patiala, Punjab.

Message from the Pro Vice- Chancellor



It is with great pleasure that I welcome you to this landmark conference on “Transforming Higher Education: Education 4.0.” This gathering marks a shared commitment to reimagining the future of higher learning at a time when education stands at the cusp of profound transformation. Education 4.0 represents far more than a technological shift. It calls for a comprehensive rethinking of pedagogy, the integration of digital fluency, and the preparation of students to thrive in an increasingly dynamic global environment.

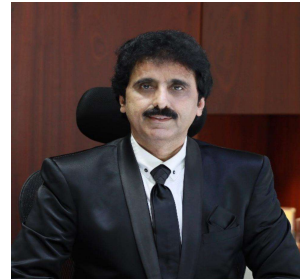
The abstracts compiled in this publication highlight pioneering research, creative solutions, and forward-looking ideas that will shape the next generation of educational institutions. The contributions from our faculty members, researchers, and academic partners span critical areas such as artificial intelligence in learning, personalized education pathways, sustainable campus initiatives, and collaborative models bridging academia and industry. Collectively, these perspectives underscore our institution’s commitment to academic excellence and to leading the transformation of higher education. I extend my sincere gratitude to all participants, presenters, and members of the organizing committee whose dedication has made this event possible. May the discussions and collaborations fostered here serve as a catalyst for meaningful and lasting change in higher education worldwide.

Prof. Ajay Batish

Pro Vice Chancellor

Thapar Institute of Engineering and Technology, Patiala, Punjab

Message from Er. Prem Kapoor
Managing Director, Shivpra Cranes Pvt. Ltd.
| BE (Mechanical),
Thapar Institute of Engineering & Technology



It is a privilege to share my wishes for the International Conference on “Transforming Higher Education for Future: Education 4.0” (September 26–27, 2025) at my alma mater, Thapar Institute of Engineering & Technology, in collaboration with NITTTR Chandigarh. The conference focuses on AI-enabled pedagogy, outcome-based education, industry–academia partnership, and employability, which is exactly what Indian higher education needs to build a workforce ready for tomorrow’s factories, labs, and startups.

From the shop floor to the boardroom, we see every day that skills, safety, speed, and sustainability now move together. Industry 4.0 has already redefined cranes, material handling, and industrial automation—embedding sensors, data, and predictive maintenance into the heart of manufacturing. Education 4.0 must therefore nurture graduates who are hands-on builders and systems thinkers—comfortable with design, data, and duty. Internships, live projects, apprenticeships, and micro-credentials tied to real problem statements will be the bridge between classrooms and competitive enterprises.

I urge students to cultivate three habits:

- Learn by doing—treat every lab and project as a prototype for industry.
- Master the basics—mechanics, controls, quality, and ethics never go out of fashion.
- Own the outcome—jobs follow those who take responsibility for results.

My compliments to the organizers, faculty, speakers, and delegates for curating tracks that span experiential learning, AI in education, advanced classroom technologies, and global accreditation. May this forum spark collaborations that place Thapar and India at the forefront of reliable, world-class, and green manufacturing.

With pride and gratitude, I wish THEC 2025 a resounding success and look forward to seeing its ideas converted into prototypes, patents, and productive careers.

Prem Kapoor
Managing Director,
Shivpra Cranes Pvt. Ltd. (TIET Alumnus)

Student Engagement, Employability and Career Readiness

Integrating Technology-based Communication Training into Engineering Education for Employability and Career Readiness

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Abstract

The engineering job market is evolving rapidly, with communication, professionalism, and career readiness emerging as core competencies expected by employers (NACE). However, engineering graduates continue to enter the workforce lacking key communication competencies, including active listening as well as oral and written communication skills. In response, accreditation bodies such as the ABET Engineering Criteria in the United States and the European Network for Accreditation of Engineering Education have recognized the need for the integration of communication training into engineering curricula as a critical and desirable educational outcome. India produces approximately 1.5 million engineering graduates annually, and only 10% are expected to secure employment in the fiscal year.

This alarming gap underscores a broader, global concern regarding the misalignment between technical education and industry expectations, particularly in communication skills, which are often overlooked in engineering curricula. In fact, industry representatives have found that 52% of engineering graduates lack adequate communication skills (Donnell & Alley). According to the NACE Job Outlook Report (2025), 77.1% of employers prioritize communication skills as a top hiring criterion, yet only 53.5% of graduates demonstrate proficiency in this area. To bridge this gap, structured and sustained communication training have become essential to meaningfully develop communication competencies in engineering students.

Unlike short-term workshops typically conducted by university placement cells, we have designed and implemented a unique one-credit Business Communication course tailored specifically for BTech engineering students. It focused on four core areas essential for employability: delivering a professional self-introduction, presenting a technical project to both technical and non-technical audiences, demonstrating global communication competencies, and preparing for interviews. The uniqueness of the course was that it employed a technology-based pedagogy that integrated AV technology in the learning experiences, as well as for assessments and tracking progress.

The key features of the technology-based pedagogy were (a) Live-recorded presentations, (b) Video-based self-assessment, (c) Longitudinal progress tracking, using archived videos, and (d) Structured tutor feedback on the video recordings. By embedding communication training within the formal academic structure and leveraging technology-based pedagogy, the course prepares engineering students for the evolving demands of the job market. Our findings suggest that the technology-based pedagogy enhanced the course engagement of engineering students and effectively trained them for the job market.

Key Words: Professional Communication, Business Communication, Engineering Education, Employability, Communication Competencies

Predictors of Academic Stress: Resilience, Coping Styles and Academic Dishonesty

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Abstract

This research explores the ways in which perceived academic stress, coping strategies, academic dishonesty, self-efficacy, and resilience interrelate to affect the academic conduct and A cross-sectional survey methodology was utilized, employing validated instruments to assess perceived academic stress, coping mechanisms, academic dishonesty, self-efficacy, and resilience Gender differences were found in two coping styles namely humour and self-blame. Resilience was also found to be different between males and females. However only, resilience emerged as predictor of academic stress. The findings also revealed greater levels of academic stress to be correlated positively with maladaptive coping approaches and a rise in academic dishonesty, whereas adaptive coping, self-efficacy, and resilience were inversely related to dishonest behaviours. Furthermore, self-efficacy and resilience played a moderating role in the connection between academic stress and dishonest actions, highlighting their protective functions. These results highlight the importance of encouraging flexible coping mechanisms and psychological resilience to lessen academic dishonesty and enhance student well-being in STEM fields. This research adds to the larger conversation about academic integrity and offers practical recommendations for teachers, administrators, and mental health professionals.

Keywords: Academic stress, coping styles, academic dishonesty, self-efficacy, resilience, STEM students

Opportunities and Challenges in Teaching Chemistry to Engineering Students in India

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Abstract

Teaching science to engineering students is necessary to develop critical thinking ability among students so that they can apply fundamental scientific knowledge in developing useful products for real-life applications and develop the skills in solving the issues faced by humanity. Specifically, the chemistry stream always remains at the forefront of the progress of human civilization, and its application in engineering is touching every aspect of modern human life.

Having taught chemistry theory as well as laboratory courses for several years to the various branches of the engineering stream, I found that teaching chemistry, a subject not directly considered relevant to a bachelor's degree, is extremely challenging. Moreover, the ongoing syllabus in different Indian engineering institutes is not directly related to the diverse engineering streams and, most of the time, is a mere extension of the +2 syllabus with a few passing industrial applications. Engineering students (specifically from computers and electronics engineering) could not connect the course contents with ongoing national or global challenges or their engineering stream. Hence, these students gradually lost interest and no longer remained enthusiastic about the subject.

To address the issue, there is a need to customize the chemistry curriculum as per the specific requirements of the branch, and the same is desired by ABET. For instance, chemistry taught to the biotechnology engineering streams should remain focused on the chemistry of biomolecules, e.g., enzyme-catalyzed reactions, the structure of proteins, DNA and RNA, interactions of a drug molecule with various biomolecules, and application of spectroscopic techniques in structural identification of biomolecules, rather than chemistry of materials or synthetic polymers. Similarly, it would be more appropriate to teach the application of different software in chemistry (e.g., SMILE, ChemDraw, Gaussian, etc.), the chemistry of organic light-emitting diodes (OLEDs), and optical fibers, and the application of AI and ML in chemistry to computer and electronic engineers. Mechanical, electrical, and civil engineers should also be exposed to nuclear chemistry, basic nuclear reactor design, electricity generation by nuclear energy, corrosion and its prevention, and other relevant topics. To sensitize the students about the energy crisis and global warming, renewable and non-conventional energy topics must be taught across the branches.

A chemistry lab is an exciting space for students to experience experiential learning and understand the practical application of chemistry in day-to-day life. Lab experiments directly related to real-life applications should be designed, such as the synthesis of biodegradable polymers, CO₂ sequestration, application of reverse osmosis in water purification, nano-scale synthesis of materials, etc., to display the problem-solving ability of chemistry.

Keywords: Chemistry teaching, Engineering streams, Chemistry lab

Teaching Engineering Drawing Through Tangible and Virtual Models: Insights from Student Perceptions

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Abstract

Engineering Drawing (ED) is a mandatory first-year subject in the undergraduate (UG) curriculum across all engineering branches at TIET Patiala. The course relies on effective visualization to foster spatial understanding. Integrating diverse model types can enhance learning by catering to varied learning styles. This study investigates student perceptions of three model types—tangible models (cardboard, polystyrene), software-based 3D models (created using Blender), and online models (from Onshape)—used as teaching aids in ED course (UES-101) during Sept-Dec 2024 semester for first-year UG students. Evaluating student perception is essential in the context of outcome-based education, as it helps align teaching methods with learner needs and expected competencies. The study involved about 130 UG students who engaged with these models across multiple topics of the course. At the semester's end, an anonymous questionnaire collected feedback on the students' perception of the models' usefulness and engagement. There were 18 questions classified into five categories: visualization, understanding of concept, engagement, skill development, and overall learning experience. Responses were acquired on a 1 to 5 Likert scale (5 being very satisfied). Findings reveal that students perceived the multi-modal approach as valuable with average ratings ranging from 4.4 to 4.7 across the 18 questions. On an average, 89% of the students rated either 4 or 5 to the questions. Across the five categories, all average ratings were above 4.5 with 'overall learning experience' and 'visualization' receiving the highest ratings of 4.7 and 4.6, respectively. The findings highlight the value of incorporating diverse model types based on student perception to enhance teaching effectiveness in ED course.

Keywords: Engineering drawing; tangible models; student perception; visualization; engineering education

Redesigning High Voltage Engineering Course Using Fink's Integrated Backward Design Process: A Case Study

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Abstract

This paper presents a case study on redesigning an undergraduate (UG) course, High Voltage Engineering, in the Bachelor of Electrical Engineering curriculum of Thapar Institute of Engineering and Technology using Fink's Integrated Backward Design (IBD) Process. The novelty of this research is associated with the application of the IBD process, considering situational factors that may affect the students' performance. The said course, which is considered a professional core course in the UG Electrical Engineering Curriculum, involves the selection and application of various types of high-voltage insulations, broadly known as insulation coordination, in the electrical power systems protection and human safety as a prime concern. Hence, a "Big Dream" is proposed for this course to encourage the students' long-term learning goals so that they will be able to design, develop, and use high voltage systems whenever they get an opportunity as an electrical engineer in their profession, maintaining optimum insulation coordination as per safety standards, even after several years of graduation. Framing course learning goals based on the "Big Dream" following Fink's Taxonomy of significant learning is considered another novel contribution of this research. The High Voltage Engineering course was taught by the author to third-year Electrical Engineering students during the odd semesters of the academic years 2022-2023 and 2023-2024. In the latter year, a thorough analysis was conducted to assess the impact of recognizing situational factors causing improvement in students' performance. The proposed approach may be implemented for redesigning other courses in higher engineering education, aiming for significant learning.

Keywords: Fink's IBD, Course Redesign, High Voltage Engineering, Big Dream, Situational Factors

Cultural Context, Stigma, and Teaching 4.0: Exploring Mediated Pathways to Academic Performance in Indian Higher Education

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Abstract

Academic performance in higher education is shaped by individual aptitude and broader sociocultural and institutional factors. In collectivist societies such as India, the cultural imperative for academic excellence often leads to heightened psychological pressure, especially when students perceive themselves as falling short of expected standards. This study explores how cultural context influences academic performance through the mediating roles of perceived stigma, internalised stigma, and Teaching 4.0, an emerging paradigm that integrates technology, personalisation, and innovation in learning. While stigma has long been linked to academic disengagement and emotional distress, teaching 4.0 presents a new avenue for re-engagement and resilience. Drawing from a sample of students in India, the study uncovers how cultural expectations, psychological burdens, and technological empowerment influence student outcomes. The findings offer valuable insights for educators and policymakers seeking to improve academic performance through culturally informed and technologically adaptive strategies.

Keywords: Cultural Context, Perceived Stigma, Internalised Stigma, Teaching 4.0, Academic Performance, Higher Education, PLS-SEM

Lifelong Learning in the Age of Digital Sociology: Navigating Echo Chambers and AI Manipulation

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Abstract

As digital platforms and artificial intelligence become deeply embedded in daily life, lifelong learning faces new opportunities and challenges. Social media echo chambers and AI-driven political manipulation shape individual behaviour, public discourse, and the skills needed for meaningful, continuous learning in the 21st century. Social media platforms like Facebook, Instagram, and (Twitter) rely on algorithms that personalize content based on user preferences. While this makes online engagement more appealing, it also creates echo chambers spaces where people are mainly exposed to ideas they already agree with. Such environments limit critical thinking, reduce exposure to diverse viewpoints, and amplify polarization. Addressing this demands that digital literacy be central to lifelong learning, equipping people to question online information, identify bias, and actively seek balanced perspectives. Meanwhile, AI and big data have revolutionized political campaigns, enabling micro-targeted ads, deepfake videos, and sophisticated influence strategies. These practices raise ethical concerns about privacy, misinformation, and manipulation. Real-world events such as the Cambridge Analytica scandal and deepfake usage in global elections show how easily AI can distort democratic processes. To respond, citizens must develop the awareness and critical skills to detect digital manipulation and protect the integrity of public debate. Here we argue that lifelong learning must evolve beyond technical skills to include critical digital literacy, ethical awareness, and civic responsibility. Drawing on insights from sociology, political science, and media studies, it outlines solutions such as integrating digital literacy into education, reforming algorithm design for diverse exposure, and regulating AI in political communication.

By rooting lifelong learning in the humanities, we can help individuals question, contextualize, and humanize technology's impact ensuring they remain informed, thoughtful participants in an increasingly digital society.

Exploring the Relationship between Personality, Academic Stress & Impostor Syndrome in University Students

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Abstract

Impostor syndrome is a psychological phenomenon experienced by high-achieving individuals who fail to internalize their success and fear being exposed as inadequately skilled. They feel they are not worthy of their success and attribute it to external factors such as luck or external circumstances. Individuals experiencing a higher level of impostor syndrome have been reported to feel underconfident as working professionals. As a result, they refrain from taking career-related risks that could boost their career growth. Therefore, it is important to recognize the factors that lead to impostor feelings in our student population for their well-being and the progress of our country's workforce. This study explores the relationship between personality, academic stress, and impostor syndrome in university students. The sample was collected from STEM students (N=126) and non-STEM students (N=43) between the ages of 18-23 for this study (N=169). A virtual questionnaire was utilised for this study, which contained the Clance Impostor Phenomenon Scale (CIPS), Mini-IPIP-15, and Perceived Academic Stress scale (PAS). The results revealed a significant positive correlation between academic stress and impostor syndrome. The personality dimensions of neuroticism and extraversion significantly related to impostor syndrome and academic stress. Gender differences were perceived in personality dimensions where female students reported higher levels of neuroticism. Gender differences for academic stress and impostor syndrome were insignificant. Similarly, STEM and non-STEM students did not show significant differences in their academic stress and impostor syndrome. Differences in STEM and non-STEM students were perceived in the personality dimensions of openness.

Keywords: Impostor syndrome, Academic Stress, Personality, STEM, non-STEM

Understanding Procrastination Patterns among School and University Teachers: A Comparative Study

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Abstract

The teaching profession is complex and demanding, with competing demands on teachers' time and attention. Teachers are responsible for preparing lessons, curriculum development, grading, classroom management, and promoting student engagement and learning. Teachers are frequently asked to carry out other core duties, such as managing professional development requirements and various extracurricular responsibilities. The purpose of the present study is to explore the prevalence of procrastination in educators with the help of a mixed-method study. In the first phase of the study a quantitative method was used to collect the data on 106 teachers, amongst whom 49 were university teachers and 57 were employed in schools. Mann-Whitney U test revealed that University teachers tend to procrastinate more than school teachers. To explore the factors behind procrastination, qualitative methods were used, and teachers were interviewed. Thematic analysis was done, and four major themes emerged: explaining the nature of procrastination, Factors leading to procrastination, Impact of procrastination, and differentiating procrastination among university and school teachers. The Present Study concludes that there exists a difference between state and trait procrastination. Several factors that lead to procrastination include workload, age, motivation level, and the nature of the task. Procrastination also directly and indirectly impacts the working and personal lives of teachers. The findings of this study will help in getting an in-depth understanding of the major causes of procrastination and will help in the development of friendly policies to manage procrastination tendencies.

Keywords: Procrastination, University Teachers, School Teachers, Mixed method, Comparative study

Project-Based Learning

Student-Led Projects in Technical Education: Opportunities and Challenges

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Abstract

In the current competitive world, learning alone cannot be considered sufficient for students graduating from technical courses. It is necessary for the students to know how the knowledge gained from a specific subject can be used in real life. The practical benefits of learning the subject should be known to the students. To immediately see the outcome of the subjects and to know the level of understanding, projects can be introduced in some subjects in the end of the semester. These projects can be specific or can be open ended. These projects can be implemented in the practical subjects such as those related to manufacturing. Open ended projects can benefit in different way as students have various challenges like selection, execution and documentation of the project. Apart from technical knowledge, such projects can enhance various capabilities of students like brainstorming, teamwork, report writing which are essential for lifelong learning. Hence, this paper provides insight into the benefits and challenges faced while inculcating open ended project specifically in Manufacturing Subject.

Keywords: Project based learning, Teamwork, Report writing

Intending Academic Excellence through Collaborative Research

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Abstract

The paper revolves around the investigations related to exploring academic excellence through collaborative research. The task was identified and taken up as the challenge by the author of the paper to explore the possibilities of collaborative research in an academic setup. The author joined hands with his collaborators in and outside his parent institute, and started working in their common areas of interest in the field of renewable energy systems and its optimization. Once the team members were confident, developed mutual trust, ready to assist and support each other empathetically, and started moving ahead with their research, the author decided to expose the post graduate students about his research in the lecture and tutorial classes. Out of the whole class of around 12 students, two students who were really excited with this field came forward, and decided to carry out their dissertation work in this field. Even the students who did not pursue their dissertation in the same field gave positive feedback, and thanked their teacher to give them the exposure about this research. The post graduate students when worked in collaboration with the author and his collaborators did wonderful job and came up with number of SCI/SCIE publications that not only improved the research profile of the institute but also motivated other students to do something good if not great. These types of initiatives can altogether bring the whole paradigm shift towards the positive growth of the institute and its stakeholder's mindset. In the end, it can be safely concluded that exploring the new possibilities of collaborative research is the need and responsibility of the present higher education system.

Keywords: Collaboration, Academic Excellence, Research Integrated Teaching, Collaborative Research

Evolution, Efficacy and importance of Capstone Projects: A Student-Centric Analysis of Design, Resources, and Collaborative Models

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Abstract

This paper presents an in-depth analysis of the evolution and effectiveness of capstone projects. It investigates how these projects have adapted over time to enhance the teaching and learning experience for both faculty and students. Through a comprehensive survey and data analysis, this research explores student feedback on the current capstone project structure, identifies areas for improvement, and proposes novel ideas for future enhancements. The study highlights the significance of capstone projects in fostering critical thinking, problem-solving, teamwork, and communication skills among students, preparing them for real-world challenges. The findings contribute to a better understanding of successful capstone project implementation and provide valuable insights for curriculum development in higher education.

Real-World Learning: A Framework for Industry-Driven Project-Based Education in Engineering

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Abstract

This paper presents a comprehensive framework for implementing industry-academia collaboration in Project-Based Learning (PBL) to strengthen engineering education through real-world engagement. Although it is good at imparting theoretical knowledge, traditional classroom training frequently lacks the experience component needed to foster problem-solving, teamwork, and industry-relevant skills. Students are exposed to real-world problems, instruments, and professional standards used in engineering practice through the incorporation of industry-defined projects within the academic program.

Throughout the project lifecycle—from problem identification to design, implementation, review, and final evaluation—faculty members and professionals from the industry collaborate to mentor the suggested model. Examples from the fields of computer, mechanical, electrical, and civil engineering show how this method improves student learning results. Students who worked on projects directed by the industry showed enhanced critical thinking, communication, creativity, and flexibility. Industry mentors enhanced the learning process and helped close the gap between academic and professional expectations by offering timely feedback and domain-specific insights. Student reflections revealed heightened motivation, greater clarity in career goals, and increased confidence in applying theoretical knowledge to practical scenarios. Industry partners also benefited by gaining access to a fresh talent pool and receiving creative inputs into real-world problems. Data from student and mentor feedback showed that over 90% of participants perceived a significant improvement in professional readiness and collaboration skills.

This paper recommends strategic alignment between institutional academic schedules and industry timelines, clear role definition for mentors, and the development of standardized evaluation rubrics co-created by both academic and industry stakeholders. The findings support the view that well-structured industry-academia collaboration in PBL can serve as a scalable model for enhancing the relevance, quality, and impact of engineering education in a rapidly evolving technological landscape.

Keywords: Industry-Academia Collaboration, Project-Based Learning, Engineering Education, Experiential Learning, Professional Skills

Enhancing Digital System Design Education through Project-Based Learning and FPGA Tools

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Abstract

Digital System Design (DSD) is a core subject in the curriculum of Electronics Engineering and related branches. Many students find the course work challenging due to its theoretical approach and lab work a bit monotonous and hence boring. To make the subject more practical, engaging and interesting, a project-based learning (PBL) model was introduced two years back for the said course in the second-year undergraduate curriculum. This approach allowed students to apply digital design concepts using a Hardware Description Language (HDL) named Verilog and Xilinx Vivado, followed by real-time FPGA implementation. With annual participation exceeding 300 students, the initiative demonstrated a consistent improvement in student performance of both the program groups, rising from an average of 7.0/10 to 7.5/10 across two years. In addition to studying textbooks, students actively engaged in coding techniques, modeling the logic blocks, simulating, and testing their designs on FPGA boards, gaining valuable hands-on experience. This shift helped bridge the gap between theory and real-world application, improving technical skills, confidence, and subject interest. By integrating innovation, self-learning, and real-world application, the model effectively embodies the core principles of Education 4.0 in engineering education, reducing the gap between industry and academia.

Keywords: Project-Based Learning, HDL, Verilog, FPGA, Xilinx Vivado, Digital Design Education.

Empowering Engineering Education through PBL: Deep Learning for Motor Imagery Using EEG Signals

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Abstract

Project-based learning (PBL) fosters experiential education by bridging theory and practice through real-world applications. This paper presents a longitudinal case study on the PBL initiative titled "Predicting Arm Movements Using EEG-Based Motor Imagery via Deep Learning." Conducted over three academic years, the initiative engaged undergraduate students in developing AI-driven models to decode motor imagery from EEG signals, enabling applications in brain-computer interfaces (BCIs), neuroprosthetics, and rehabilitation. Each year, three student teams per batch undertook the end-to-end pipeline—EEG data acquisition, signal pre-processing, feature engineering, and model training. Deep learning architectures, including CNNs, LSTMs, and hybrid models, were iteratively explored and optimized for classifying imagined arm movements. Despite achieving steady improvements in prediction accuracy and real-time deployment (e.g., robotic arm control, VR simulations), challenges such as EEG noise, subject variability, limited datasets, and model generalization persisted. Addressing these hurdles required adaptive pre-processing strategies, careful model validation, and collaborative problem-solving. The iterative process resulted in higher prediction accuracy, robust models, and improved real-time adaptability. Students gained interdisciplinary expertise across neuroscience, AI, and rehabilitation engineering, fostering innovation-driven research. The project contributes to next-generation neuroprosthetics and adaptive assistive systems. Key outcomes include the development of robust classifiers, improved temporal resolution, and deployment-ready prototypes. Beyond technical achievements, students reported enhanced understanding of biomedical signal processing, interdisciplinary teamwork, and research mindset. The case study underscores PBL's potential in training future-ready engineers, advancing neurotechnology, and instilling innovation through authentic, challenge-driven learning experiences.

Towards Successful Integration of Project-Based Learning in Higher Education: Challenges, Enabling Technologies, and Implementation Strategies

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Abstract

Project-Based Learning (PBL) is a dynamic, student-centred approach to education where learning happens through actively exploring real-world problems and challenges related to the course. In this approach, instead of passively absorbing information about the technical concepts, the students engage in projects, with real life applications. This requires critical thinking, collaboration, and creativity over an extended period of the course and sometimes outside the classroom also. The integration of Project-Based Learning (PBL) in higher education holds immense potential to enhance student engagement, critical thinking, and real-world problem-solving skills. However, successful implementation remains a challenge due to institutional inertia, faculty readiness, and infrastructural limitations. This paper explores the multifaceted challenges institutions face while adopting PBL, the technological tools that enable effective project-oriented instruction, and the pedagogical strategies that drive meaningful outcomes. Drawing on case studies and current research, this article proposes a structured framework for embedding PBL into diverse academic contexts, emphasizing the importance of cross-disciplinary collaboration, formative assessment, and continuous feedback. The study aims to guide educators, administrators, and policymakers toward fostering a sustainable and impactful PBL ecosystem in higher education.

Enhancing Practical Learning through Project-Based Learning: A Classroom Intervention Study

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Abstract

Project-Based Learning (PjBL) has emerged as a powerful pedagogical approach to bridge the gap between theoretical instruction and real-world application. This study investigates the implementation and impact of PjBL in a second-year undergraduate Artificial Intelligence (AI) course. Initially employing traditional problem-based learning, students struggled to translate algorithmic knowledge into practical implementations. A classroom intervention introduced PjBL, requiring student teams to design and develop AI projects based on real-world problems without using predefined APIs or libraries. Results from direct and indirect assessments show significant improvement in student performance, engagement, and understanding of practical AI. This article contributes empirical evidence supporting PjBL as a robust strategy for AI education.

Keywords: Project-Based Learning, Artificial Intelligence Education, Engineering Pedagogy, Practical Learning, Classroom Intervention

Enhancing Students' Understanding of Manual and Automation Testing Through Project-Based Learning

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Abstract

Project-Based Learning (PBL) has emerged as a transformative pedagogical strategy in engineering education, particularly in bridging the gap between theoretical knowledge and practical application. This paper presents a research study implemented within the Test Automation Lab course for undergraduate engineering students, focusing on enhancing their understanding of manual and automation software testing. The intervention involved structured project-based activities conducted over a 15-week semester, where students applied testing tools like Selenium and TestNG in team-based real-world scenarios. Through a blend of hands-on tasks, peer learning, and continuous feedback, the initiative aimed to develop critical thinking, problem-solving, and industry readiness. Evaluation was based on survey feedback, mid-semester reviews, and final assessments, revealing that over 80\% of students reported improved conceptual knowledge and application skills. The paper contributes to educational research by demonstrating the efficacy of PBL in software testing education and offers insights into instructional design, assessment strategies, and learner engagement in technical disciplines.

Keywords: Project-based learning, Software Testing, Peer-to-peer learning, Selenium, TestNG.

Project-Based Learning in Engineering Physics: A Reflective Pedagogical Intervention

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Abstract

This study offers a reflective account of the implementation of a Project-Based Learning (PBL) strategy within the Applied Physics course (UPH013) for first-year undergraduate engineering students at Thapar Institute of Engineering & Technology. Driven by the pedagogical challenges presented by teaching abstract and mathematically rigorous topics—such as quantum mechanical wave functions, diffraction patterns, and probability densities—this intervention sought to enhance conceptual understanding through active, student-centered engagement.

The PBL module, which lasted 7–8 weeks, was seamlessly integrated into both theoretical and laboratory components, involving the creation of interactive, Python-based visualization projects. Students collaborated to explore selected topics by building models that facilitated real-time parameter manipulation and peer demonstrations.

Feedback from 67 students indicated significant improvements in comprehension, critical thinking, collaborative learning, and sustained interest in the subject matter. Reflecting on the experience, this initiative showcased the transformative potential of PBL in fostering learner autonomy, meaningful assessment, and engagement. While some logistical and collaborative challenges emerged, the experience underscores the importance of incorporating experiential learning approaches in STEM education to align with evolving pedagogical paradigms and institutional objectives.

Keywords: Project-Based Learning, Conceptual Understanding, Active Learning Strategies, Physics Education

Global Implications of Education

Vocational Education and Training as a Tool of Skill Development in India: A Critical Analysis

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CONTEXT

Vocational education and training prepare learners for the job and make them more effective by giving an opportunity to participate and learn from firsthand experiences. National Education Policy (2019) suggested including vocational education into mainstream education by revamping all the aspects of vocational education and training. Despite the importance of vocational education and training, there are many challenges associated with it.

PURPOSE OR GOAL

The present study will inspect some of the crucial challenges of vocational education and training. The present study will also try to find the future improvement and development of vocational education and training. The study will throw some light on the cope and challenges of the National Skill Development Policy (NSDP). Last but not the least the study will suggest some strategies to enhance employability skills in order to increase competitive advantage in the long run.

ACTUAL OR ANTICIPATED OUTCOMES

The study reveals some key challenges of vocational education and training especially in India. Some of the vital challenges are lack of proper institutions, lack of proper attitude among people, scarcity of good teachers etc. In order to deal with the challenges setting up more institutions to impact vocational training and education is required. In addition to this the vocational education should not be separated from mainstream education rather it needs to be included mandatorily in mainstream education in order to increase competitive advantage. Skill development via vocational education and training is the most important aspect in the development of a country, hence, it needs coordinated efforts by the government and other stakeholders.

CONCLUSIONS/RECOMMENDATIONS/SUMMARY

Vocational education aims to empower individuals to take up employment in a skilled craft or workforce and a skilled workforce is crucial component of education for sustainable development (ESD). It often goes hand in hand with career and technical education. Issues with implementation, organization facilitation, lack of experts in the field, and social stigma acts as significant barrier to the growth and development of VET in India. But with increased opportunities and optimum utilization of Vocational educators and trainers, India can upscale its reach with skilled manpower and create worldwide importance of the same.

Keywords: Vocational education and training, NEP 2019, self-realization, skill development, education for sustainable development.

Breaking Cultural Barriers in Education: MOSAIC's Multimodal AI for Cross-Cultural Learning and Heritage Preservation

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Abstract

Cross-Cultural Communication faces extraordinary challenges as digital globalisation is at its peak, yet existing large language models (LLMs) exhibit significant Western cultural bias and lack multimodal cultural understanding capabilities. Current translation systems fail to preserve cultural subtlety and contextual significance when processing combined text and visual cultural content. We present MOSAIC (Multimodal Ontological System for Adaptive Intercultural Communication), a novel multimodal framework that integrates culturally aware text processing with visual cultural understanding to enable authentic cross-cultural translation and heritage preservation. Our approach combines multilingual language models with vision transformers, unified through a cultural knowledge graph to enable culturally aware cross-modal understanding.

The system employs hierarchical cross-modal attention mechanisms and cultural context classifiers to generate culturally suitable explanations that maintain both factual accuracy and cultural sensitivity. We introduce a contrastive learning framework that aligns cultural concepts across text and visual modalities, enabling the system to process combined cultural artifacts and generate contextually rich explanations. Our framework demonstrates promising improvements in cultural translation and cross-modal understanding on diverse heritage datasets. Preliminary evaluations indicate enhanced translation accuracy, strong alignment between text and imagery, and high cultural appropriateness ratings across multiple contexts. In educational settings, our approach effectively supports cross-cultural comprehension, with initial user studies reflecting high levels of engagement and cultural appropriateness. This work contributes the first multimodal culturally aware translation system and provides a foundation for AI-assisted cultural preservation and inclusive digital communication.

Keywords: Multimodal AI, Cross-Cultural Learning, Cultural Heritage Preservation, Educational Technology, Large Language Models

Reimagining Engineering Education in India: A Strategic Framework for Quality Enhancement

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Abstract

Recent developments, such as the introduction of the National Education Policy (NEP) 2020, greater emphasis on outcome-based education (OBE), and the integration of emerging technologies like AI, data science, and sustainability practices, are reshaping engineering education in India. The present study develops a model for Augmenting the Quality of Higher Education (AQHE) by identifying critical success factors such as academic culture, student engagement, employability, and faculty involvement. academic culture, academic environment, academic operations, student engagement, learning assessment, student satisfaction, employability of students, employee involvement, employee attraction and transnational branding of institutions. This model provides strategies to enhance institutional performance and ensure engineers are equipped for modern challenges.

Quality in higher education is a multidimensional construct encompassing academic standards, faculty expertise, curriculum design, research output, and student outcomes. In the context of engineering education, quality assumes greater significance as engineers contribute directly to innovation, employability, and national growth. Despite India's rapid expansion of engineering institutions, persistent challenges such as outdated curricula, inadequate industry integration, insufficient research culture, and uneven faculty training undermine the effectiveness of higher education institutions (HEIs).

Faculty Development and Support

Generativity Theory: Potential to Transform Learning Pedagogy for Sustainable Entrepreneurship

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Abstract

Generativity Theory has been observed as a contributor to enhancing employee creativity and in turn help businesses achieve superior results. In context of sustainable entrepreneurship, it has been observed as leading to multi layered innovation while being aligned to the evolution of future of work. Conceptually, the theory is technological in nature. However, when integrated or linked to the Theory of Effectuation which is process oriented – the potential to deliver on Sustainable Development Goals or inculcating a culture of Sustainable Entrepreneurship increases manifold.

The creativity competencies of Generativity Theory hold potential to be included, deeply referenced in curriculums of business or liberal arts schools thus transforming current pedagogies to nurture and train future managers. These young managers could then enter the corporate workforce with a mindset which is rooted in creativity, oriented to Sustainable Goals, Priorities, and/or Entrepreneurship with the goal of success in business metrics. Initiatives like UN-PRME are playing a key role in this effort and when leveraged with end-to-end linkages, could further increase progress. Global Consulting firm BCG has formalized, researched and done case studies with organizations with to tackle multiple challenges with the concept of Generative Leadership principles.

An Intersectional Review of Theories of Generativity (including its Creativity Competencies), Effectuation and UN-PRME principles could hold long term potential for transforming learning pedagogies at Institutions, thus skilling talent to manage challenges of the Future of Work. Education 4.0 is thus a key enabler for success of Industrial Revolution 4.0.

Keywords: Creativity, Effectuation, Generativity Theory, Learning Pedagogy, Sustainable Entrepreneurship, Transformation

Skill Enhancement Courses Under NEP-2020: A Training Module for First Responders to Crime

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Abstract

The foremost official to reach a scene of crime is a policeman, who is designated as First Responder to Crime or First Officer. In India, the term Investigating Officer too is quite common. At the initial stage, the crime scene is ridden with chaos and confusion. It is the duty of the First Responder to prevent the chaos from being destructive to the site, as well as to the crime evidence dispersed therein. The decisions of the officer bear a strong influence on the eventual outcome of the criminal investigation. It is therefore pertinent that she/he must receive training and guidance to inculcate the spirit of scientific temperament in decision making. For this reason, the discipline of Forensic Science has been inducted into NEP-2020 under the category of Skill Enhancement Courses. At many institutions, the law enforcement officials are permitted to enroll for the forensic science course, irrespective of their previous academic background.

This presentation highlights the need for designing a module for the training of the First Responders to Crime so as to strengthen the criminal justice system in India.

Reflective Teaching as a Practical Approach to achieve Academic Excellence

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Abstract

This paper aims to investigate some real insights of the power of reflective teaching – a process of self-observation and self-evaluation. The whole teaching learning process requires thoughtful preparation and effective execution of the planned lectures. One needs to understand that teaching and learning are the mirror images of each other where in today's modern era, students and teachers need to be considered as partners and facilitators, respectively, in the whole teaching learning paradigm. For effective engagement with the students, the role of reflection whether in action or action, needs to be understood. To make the reflective practices an integral part of one's teaching, one must understand the importance of timely and effective feedback and assessment tools. Teachers definitely get an opportunity to identify areas for improvement, adapt their strategies, and ultimately create an active learning environment for the students ensuring that they remain responsive to the dynamic needs of their students. The author highlights various reflective practices, such as peer-to-peer learning, peer observation of teaching and learning, journaling, experiential work, collaborative tasks, and suitable delivery and assessment techniques highlighting their applicability across varied domains. The findings suggest that reflective teaching has the potential to make the positive cultural shift of continuous improvement in the modern education system ultimately benefiting all the direct stakeholders and the society as a whole.

Exploring the Impact of Digitalization on Teachers' Effective Communication

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Abstract

Quality Education is among the 17 Sustainable Development Goals (SDGs) established by the United Nations within the framework of the 2030 Agenda for Sustainable Development. A high-quality education depends significantly on effective educators who can cultivate stimulating learning atmospheres, promote analytical thinking, and give tailored support to the learners. Loy (2006) highlights that educators who possess effective communication skills are more capable of cultivating a supportive learning atmosphere, enhancing student participation, and ultimately leading to better student results. In the digital age, effective teacher communication includes both digital tools like social media, online platforms, and multimedia, as well as more conventional abilities like clarity, active listening, and positive body language. As education becomes increasingly digital, it is essential to organize educational content through technological advancements (Arisoy, 2022). The current study seeks to gain insight into how digital technology impacts effective communication of the teachers, providing innovative tools and platforms that improve the extent and quality of interactions in educational environments. The digital transformation has significantly changed the way educators engage and interact with their students, as it provides various advantages, including enhanced accessibility and innovative teaching approaches. Nonetheless, it also requires careful management of issues such as miscommunications in virtual settings and concerns regarding data security.

Keywords: effective communication, digital technology, innovative strategies, online teaching, miscommunication

Effectiveness of Teaching Tools Used in Student-Centred Learning

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Abstract

This paper stems from the participation in the New Directions in Teaching and Learning Program 2017 and Advanced CAPSL Program 2023 at Thapar Institute, which introduced innovative student-centred teaching strategies. Motivated by these techniques, a range of active learning tools such as think-pair-share, peer assessment, flip charts, one-minute papers, punctuated lectures, and muddiest points in undergraduate classroom over a period of three months has been implemented. The primary goal was to enhance student engagement, deepen conceptual understanding, and improve the overall learning experience.

To evaluate the effectiveness of these strategies, a structured feedback form has been designed consisting of six questions rated on a 10-point scale. Feedback was collected from all students and analysed to assess the impact of each tool. Results indicated strong student support for techniques like think-pair-share, peer assessment, and the real-life relevance of course content, with scores predominantly between 5 and 9. However, flip charts received lower effectiveness ratings, largely due to visibility and handwriting issues in large groups.

Further efforts included encouraging students to engage in end-of-lecture activities, group projects, and theory-to-practice connections in lab work - all of which received positive responses. Challenges included managing time constraints, especially due to student placements and the additional time required for interactive methods. Despite these limitations, the outcomes were encouraging, reflecting improved student confidence, motivation, and participation.

Supported by scholarly literature in higher education pedagogy, this paper demonstrates the potential of active learning to transform traditional teaching. Main intend is to extend these strategies across larger student groups to sustain and scale the impact.

Keywords: Think-pair-share, peer assessment, flip charts, one-minute papers, punctuated lectures, and muddiest points

Assessing the Pedagogical Approach and Relationship between Personality Traits and Communication Styles of Teachers

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Abstract

To communicate information, ideas, opinions, or feelings, whether they are impersonal or personal, it is essential to have a thorough understanding of the primary communication styles. For professionals, it is frequently challenging to identify a communication style that is particularly effective in fostering change and ensuring the recipient's gratification, and communication styles are characteristic adaptations of personality (Waldherr & Muck, 2011). The current research attempts to explore the relationship between teachers' personality and their communication style while interacting with the students in a classroom paradigm. Two instruments were administered to six hundred faculty members teaching in technical and non-technical institutions in the northern region of India, the HEXACO-PI-R model of personality (Ashton et al., 2004; Ashton & Lee, 2008), and an adapted version of the Communication Styles Inventory (CSI) by De Vries et al. (2013). Descriptive and inferential statistics were used to analyze the responses gathered from self-reporting questionnaires. The results of the Pearson correlation coefficient indicated a significant positive correlation between personality traits and communication styles of faculty. A substantial correlation was also reported between the six different domains of personality and the six domain-level communicative behavior scales of CSI. The study makes significant practical contributions as teachers' personality traits and styles have an impact on the learning environment in the classroom; hence, it is vital to consider them while conceptualizing the effectiveness of teachers, who facilitate students' class participation, academic performance, and social skills.

Keywords: Personality traits, Communication styles, Classroom teaching, Self-assessment,

Reflective Teaching as a Practical Approach to Achieve Academic Excellence

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Abstract

This paper aims to investigate some real insights of the power of reflective teaching – a process of self-observation and self-evaluation. The whole teaching learning process requires thoughtful preparation and effective execution of the planned lectures. One needs to understand that teaching and learning are the mirror images of each other where in today's modern era, students and teachers need to be considered as partners and facilitators, respectively, in the whole teaching learning paradigm. For effective engagement with the students, the role of reflection whether in action or action, needs to be understood. To make the reflective practices an integral part of one's teaching, one must understand the importance of timely and effective feedback and assessment tools. Teachers definitely get an opportunity to identify areas for improvement, adapt their strategies, and ultimately create an active learning environment for the students ensuring that they remain responsive to the dynamic needs of their students. The author highlights various reflective practices, such as peer-to-peer learning, peer observation of teaching and learning, journaling, experiential work, collaborative tasks, and suitable delivery and assessment techniques highlighting their applicability across varied domains. The findings suggest that reflective teaching has the potential to make the positive cultural shift of continuous improvement in the modern education system ultimately benefiting all the direct stakeholders and the society as a whole.

Enhancing Student Understanding in Data Structures and Algorithms through Visualization and Animation Tools: An Intervention-Based Study

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Abstract

Learning Data Structures and Algorithms (DSA) continues to be a significant challenge for undergraduate programs due to the abstract and procedural nature of algorithmic concepts. Although, there are previous researches that has found visualisation and animation tools can increase understanding, still few studies assess the impact of these tools in a controlled classroom settings. To fill this gap, a quasi-experimental, post-test control group approach is employed to investigate the effects of visualization-enhanced teaching methodology. The experiment involves the participation of a group of 175 third-year engineering students which is randomly split into either a traditional teaching group or a visualization-based group. To avoid bias, cumulative GPA are used while forming the corresponding groups. The visualization-based teaching methodology incorporates normal lectures followed by interactive platforms that allow students to visualize state changes in algorithms. The descriptive statistics of the preliminary analysis indicate that the students who used visualization and animation tools achieve comparative higher mean score and reduced variability in performance. These results contribute towards research based pedagogical approaches in computing education and supports integrating interactive visualization into core elements of the curriculum to develop richer forms of algorithmic reasoning and equitable learning opportunities for learners. Keywords Visualization Tools, Algorithmic Learning Outcomes, Educational Intervention, Interactive Learning

Emotional Landscape of Educators: Challenges to Academic Impact

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Abstract

Teaching as a profession is emotionally demanding, requiring the management of emotional challenges arising from work pressure and expectations. Teachers' emotions play a significant role in the teacher-student relationship, classroom teaching, and professional responsibilities. However, there is a dearth of research that focuses on understanding the role of emotions in classroom teaching and learning from both teachers' and students' perspectives. This qualitative study explores how teachers regulate their emotions and how these strategies influence classroom environments and students' academic experiences. Drawing on semi-structured interviews with 20 teachers [10 school and 10 university teachers] and focus group discussions with 50 university students, the research adopts a participant approach to examine the emotional dynamics of teaching. Thematic analysis (Braun & Clarke, 2006) revealed that teachers encounter substantial emotional challenges, including burnout, emotional spillover, and societal pressure to suppress emotions. However, mindfulness, emotional boundary-setting, and peer support help teachers manage these challenges. Students reported that teachers' emotional states significantly affect their engagement, confidence, and subject preferences. The findings emphasize the importance of emotionally aware teaching in fostering a positive learning atmosphere and enhancing students' academic motivation and well-being. This research highlights the critical role of teachers' emotions in education. Teachers should be trained to manage emotional burnout, stress, and workload pressure. Educational institutions and policymakers should also create an emotionally supportive environment for teachers to strengthen and support student learning.

Keywords: Emotion regulation, Emotional labor, Teacher-student relationship, Classroom climate, Qualitative research.

Role of Appropriate and Timely Assessment and Feedback in Higher Education

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Abstract

Higher education needs continuous improvement in teaching and learning pedagogy for the effective classroom engagement. The role of appropriate and timely assessment and feedback in higher education is pivotal while fostering effective student learning and academic success. Assessment and feedback strategies are vague if they are not appropriate and not done in a timely manner. This paper explores the role of appropriate assessment and its time frame for evaluation, along with bi-directional feedback. Feedback of well-structured assessment schemes, aligned with purposeful learning outcomes, not only evaluates student performance but also infuses deeper engagement along with critical thinking. Timely feedback by students, when delivered constructively, acts as a powerful pedagogical tool to improve conventional delivery and assessment methods and provides a platform to the teachers to reflect on their teaching learning process, identify gaps in assessment, and make meaningful improvements later. Feedback must be very specific, actionable, and contextualized so that its true impact can be visualised in teaching and learning pedagogy. Furthermore, the cyclic integration of formative assessments and regular feedback not only enhances motivation but also supports self-regulated learning and contributes to overall academic satisfaction. By examining current practices and challenges in assessment and feedback, this paper highlights the necessity for educators to adopt a learner-centred approach in assessment and feedback while maximizing student development and academic achievement in higher education.

Redefining Formative Assessment: A Dual Approach Using Quick Response and Attendance Patterns

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Abstract

Formative assessment is pivotal in modern engineering education, yet conventional feedback mechanisms often treat all student responses with equal weight, regardless of engagement levels. This paper proposes a novel formative assessment framework that dynamically allocates weightage to student feedback based on behavioral indicators such as responsiveness, participation, and attendance patterns. Central to this model is the *Quick Response Ranking* (QRR) approach, wherein students who actively engage during instruction—by responding promptly to in-class queries or completing lab tasks ahead of peers—receive proportionally higher influence in the feedback aggregation process. This not only promotes active learning but also offers instructors more representative insights into effective pedagogical strategies.

Additionally, the assessment model incorporates an *Irregularity Factor* (IF) to refine interpretations of student attendance. By analyzing irregularity trends and contextual factors, the IF distinguishes reluctant or disengaged learners from students facing genuine emergencies or constraints. This dual-layered scheme ensures that feedback-driven instructional adjustments are informed by genuinely engaged learners without penalizing those hindered by valid circumstances.

Preliminary implementation across multiple undergraduate engineering courses indicates that this refined feedback model enhances teaching adaptability, student involvement, and perceived fairness. The framework encourages proactive classroom behavior, fosters self-regulation, and empowers educators with nuanced, data-informed instructional insights. This paper presents the methodology, implementation strategy, and empirical analysis of the model's impact on teaching-learning outcomes, offering a scalable blueprint for holistic formative assessment in technical education.

Keywords: Formative Assessment, Quick Response Ranking, Weighted Feedback Mechanism, Student Engagement Metrics, Attendance Irregularity Factor

Financial Literacy as a Protective Factor Against Materialism - Driven Impulsive Buying

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Abstract

Traditional academic curricula frequently fall short in providing students with critical life skills, especially financial literacy, in an era of rising consumerism and rapid societal change. Many university students are using unhealthy coping strategies like impulsive buying as a result of the significant increase in consumer purchasing power in recent years, particularly among young adults. Materialistic values and a lack of sound financial knowledge are often the driving forces behind this behaviour, which leads to financial stress and long-term effects on one's personal wellbeing. The urgent need to integrate financial literacy into higher education is underscored by this emerging trend.

The purpose of this study is to investigate the connection between university students' impulsive purchasing behaviour, materialism, and financial literacy. One hundred university students will be chosen as a sample. Standardised tools will be used, including the CFPB Financial Well-Being Scale for measuring financial literacy, the Materialism Scale by Richins and Dawson (1992), and the Impulsive Buying Scale by Rook and Fisher (1995). Hayes' Process Macro will be used to analyse the data in order to look at the moderating and mediating effects of the variables.

According to the study, it is not only appropriate but also crucial to incorporate financial literacy instruction into university curricula. In order to make education more pertinent to today's societal issues, the findings seek to support experiential learning models and policy changes that encourage sustainable consumer behaviour and long-term financial security.

Keywords: Impulsive Buying, Materialism, Financial Literacy, Academic Curriculum

Experiential Learning

Redefining Engineering Curriculum Aligning with Changing Dynamics through Innovation and Entrepreneurship

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This paper presents addressing gaps in traditional engineering education through systematic integration of innovation and entrepreneurship, a seven-phase entrepreneurial development framework implemented at Sri Ramakrishna Institute of Technology (SRIT). The methodology encompasses design thinking, innovation exposure, prototype development, project evaluation, funding acquisition, academic integration, and startup support phases. Through the Hub for Innovation and Entrepreneurship (HIVE), the framework has produced commercially viable products, including precision fish-cutting technology, AI-powered medicinal plant recognition systems, and smart fishery boats. Successful startup outcomes include Shrimp Hoard Technologies (INR 33 lakhs funding), Pezray Technologies, MECHATRON MOTORS LLP, and Hussk'd. According to the results, 85% of projects achieve Technology Readiness Level 4 or higher, and government funding integration and NPTEL/IBM certifications improve student engagement. The framework offers a scalable model for engineering education that promotes Industry 4.0 readiness by showcasing enhanced employability and successful technology transfer from academia to market.

Keywords: Engineering Education, Innovation Framework, Entrepreneurship Development, Technology Transfer, Curriculum Design, Student Startups, Academic-Industry Collaboration

Augmented and Virtual Reality in Education: A Comprehensive Review of Immersive Learning under Education 4.0

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Abstract

Immersive technologies have left a great impact on many eras but in the field of education, AR, VR technologies have changed the perspective of thinking. Learning then evolves into something active and exciting and is often provided on an ad hoc basis; hence, plenty of opportunities become available owing to these technologies. This research attempts to track the AR and VR profiles in education and see how Education 4.0 principles of learner-centric approaches, intelligent learning environments, and digital skill evolution intersect with these technologies. We examined the immersive nature of education and the ways in which it is advantageous for learning through the application of technology, which also facilitates the implementation of immersive learning. Additionally, we address several of its most significant challenges, including the necessity of teacher training, the high costs of implementation, the features' limitations, and concerns regarding availability. Moreover, we report on the new trends and possible future plans such as the use of AI, haptic feedback, and mixed virtual reality in this sector. Essentially, the research is a complete view of the virtual reality technologies in the educational future through the recently published materials, case histories, and expert opinions. It even goes on to talk about the strategic matters that should be addressed for the proper use of these technologies and to have a deep impact.

Keywords: Augmented Reality (AR), Virtual Reality (VR), Education 4.0, Teacher training, Teaching simulations, AR/ VR in Education.

Rebuilding Heritage, Rebuilding Skills: A Stepwell Game Teaching Transversal Skills for Sustainability to Engineering Students in India

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Abstract

Transversal skills such as perspective-taking, negotiation, critical thinking, systems-thinking, are needed in 21st century engineers, yet are not typically taught in the current engineering curriculum in India, limiting the students' competencies to deal with real-world and interdisciplinary challenges. This paper presents a gamified intervention to teach transversal skills in classrooms. Anchored in 3T-PLAY's Trident framework¹ and adapted by Plaksha University for India's groundwater crisis, cultural history, and community-centred infrastructure, it serves as a primer for institutions aiming to integrate such skills using a gamified approach. The activity was structured across three phases: "Knowing" (background context and concepts), "Experiencing" (role-play and negotiation) and "Learning from Experience" (reflection and synthesis). The paper describes our 80 minutes in-class modules aimed at undergraduate engineering students, anchored in the historical Indian context of building/re-building a demolished stepwell. Using a gamified approach supported by LEGO bricks, students assumed one of four stakeholder roles—civil engineer, cultural consultant, contractor or village sarpanch and negotiated the design trade-offs to conform to the real-world parameters (durability, cultural relevance, cost and sustainability). The post activity reflections in the second lecture revealed that the students reported a strong need to consider different perspectives and the need for environmental preservation alongside technical feasibility. The paper highlights how reimagining heritage infrastructure through interactive pedagogies can cultivate both technical and ethical literacy. Under the Education 4.0 framework, it provides a replicable model for incorporating heritage-based learning and Education for Sustainable Development (ESD) into engineering curricula.

Keywords: Transversal skills, Stepwells, Sustainability, Experiential learning, Design Thinking

Geo-Education through Experiential Learning Approach

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Abstract

Current generation of gadget reliant students find it difficult to visualize the concepts on their own. Under the circumstances, Experiential Learning Approach has been found to be helpful. The fundamental concept of this approach relies on learning of participants as they comprehend the concepts problem statement experience in physical form. Present study discusses the advantages in the context of enhanced conceptual clarity regards slope stability, effective stress and consolidation, which are topics related to Geotechnical Engineering. Student groups are tasked to discuss and then demonstrate the stability of slope when subjected to various triggering factors. The triggering factors are selected based on an understanding of factors causing slope failure in the field. In order to achieve this, the groups construct physical models of the slope in a Tilting table. Efficiency of their suggested intervention is measured by (i) undercutting the toe, and (ii) raising the table from one side, thereby increasing the slope angle. The effect on student comprehension is quantified by comparing their performance in written examination. The study proves the effectiveness of Experiential based approach in Geo-education and the need to incorporate more such intervention in teaching pedagogy.

Keywords: Experiential Learning Approach, Geo-Education, Slope stability, Physical model, Teaching

The Indispensable Role of Digital System Design Course in Conducting Experiential Learning Activities

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Abstract

Experiential learning is a cornerstone of modern engineering pedagogy, crucial for transforming theoretical knowledge into practical competence. This paper argues for the indispensable role of the Digital System Design (DSD) course as a primary conduit for conducting effective experiential learning activities for undergraduate engineering students. By its very nature, DSD necessitates hands-on engagement with hardware description languages (HDLs) like Verilog and VHDL, simulation tools, and Field-Programmable Gate Arrays (FPGAs). We elucidate how the DSD curriculum provides the fundamental concepts—from logic gates and state machines to complex architectural design—that directly enable project-based learning. Through three diverse case studies—a digital clock with hour, minute, and second display; a camera chip development using FPGA; and a laser-based sensor system on FPGA—we demonstrate how DSD projects foster critical problem-solving, debugging, and iterative design skills. Moreover, these activities inherently promote teamwork, communication, and an appreciation for real-world design constraints. The immediate feedback from physical hardware implementation significantly deepens understanding, enhances retention, and builds student confidence, thereby preparing them effectively for the demands of the modern electronics industry.

Keywords: Experiential Learning, Digital System Design, FPGA, Verilog, VHDL.

From Policy to Practice: Implementing Experiential Learning in Higher Education under NEP 2020

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Abstract

Education plays a pivotal role in unlocking individual potential, fostering equity, and driving national development. The National Education Policy (NEP) 2020 presents a transformative vision for India's higher education system, emphasizing flexibility, inclusivity, and the integration of technology to meet the evolving needs of learners. A cornerstone of NEP 2020 is its strong focus on experiential learning, a pedagogical approach that promotes active student engagement, practical application, and critical thinking. Experiential learning is particularly significant in today's complex educational landscape, where traditional methods often fall short in addressing diverse classroom dynamics. NEP 2020 underscores the need for a paradigm shift in pedagogy to make education more holistic, inquiry-driven, competency based, and learner centered. This approach encourages both teachers and students to engage deeply with the curriculum, fostering real-world problem-solving skills and hands-on learning experiences. The successful implementation of NEP 2020 depends largely on the digital transformation and adaptability of higher education institutions. As the policy aims to prepare students for future challenges, experiential learning emerges as a key enabler in bridging the gap between theoretical knowledge and practical application. This research demonstrates the vitality of experiential learning as advocated by NEP 2020, examining its role in enhancing teaching methodologies and student learning outcomes in higher education.

Keywords: “NEP 2020, experiential learning, higher education, curriculum, problem-solving technique

“Problem Design Thinking”- A Case Study based on Experiential Approach

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Abstract

Experiential Learning, an innovative educational approach, is particularly effective at connecting theoretical knowledge with real-world application. This case study investigates the use and impact of experiential methods in postgraduate education, emphasizing how hands-on experiences enhance students' abilities in critical thinking, problem-solving, and professional readiness. The focus of the case study is on a group of postgraduate students who participated in an experiential learning program entitled "Problem Design Thinking." To evaluate student involvement, knowledge retention, and skill advancement, data was collected through performance assessments, personal interactions, continuous lab evaluations, and student reflections. The findings indicate that students' ability to utilize theoretical concepts in tackling complex real-world issues has notably enhanced, along with their skills in communication and teamwork, as well as their self-assurance in professional settings. Moreover, students expressed an increased eagerness to engage in learning and perform independent research, alongside a more profound comprehension of their discipline. Students also indicated a heightened enthusiasm for self-directed study and research along with an enhanced understanding of their academic discipline. The study further addresses the challenges faced, such as the complexities of evaluations and the need for faculty training. In summary, the case study emphasizes the importance of experiential learning as an effective tool for postgraduate education that fosters both professional and academic growth. It advocates for the integration of this approach into curriculum development to prepare students with the competencies needed to effectively handle changing, real-world situations while also gaining expertise in their respective fields. This approach positions experiential learning as a strategic educational philosophy for higher education, instead of merely a teaching method.

Keywords: experiential learning, design thinking, assessment, hands-on experience.

Scope for Transformation in Higher Education: The Possibilities for Collaborative Experiential Tasks

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Abstract

This paper intends to investigate the scope of improving the engagement level of students in higher education which advocates the inclusion of collaborative experiential tasks during the delivery and the assessment of course module. These tasks not only enhance the understanding level of the students but also improve their inter-personal relationships. It is seen that students take more interest in these types of activities rather than their boring stereo type lectures. These activities also help in managing the anxiety and stress level in students that usually accompany their written examinations. This intervention was applied by the course instructor in one of his courses to improve the physical and mental presence of students in the class. Students were also involved in the process of project identification, exploring various problem-solving methods, peer-to-peer teaching learning, role identification of each member in the group, and self and peer assessment. The involvement of students in these activities not only improved their academic performance in these evaluations but also made them more responsible after they are made aware about their specific roles. The students' feedback suggests that they really enjoyed working in groups on real-world problems. The satisfaction level of the course instructor also reached a different level seeing his students doing some serious work. So, it can be safely concluded that collaborative experiential tasks can definitely open new horizons in today's higher education system. This will not only improve the cohesiveness among all the stakeholders but also motivate all to work for seeking excellence together.

Circuit Component Identification and Layout Generation using AI

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Abstract

Printed Circuit Boards (PCBs) connect different components in electronic devices, allowing communication between them. With growing design complexity, there is a need of intelligent tools to automate and optimize PCB layouts. This experiential learning activity introduces a CNN-based system that extracts key features from existing PCB designs and uses these features to redesign the PCB layout. Existing Electronic Design Automation (EDA) tools offer limited support for feature extraction from PCBs. The aim is to address this gap by developing a deep learning-based system that can extract meaningful features from PCB layouts using Convolutional Neural Networks (CNNs) and apply optimization techniques. The activity follows a structured approach, beginning with collecting and annotating PCB layout images and schematic files working hands on experience. These are pre-processed using pyTorch. Then, a CNN is trained to extract critical features. The extracted features are represented through graphs and used to guide automated reconfiguration of the PCB layout using algorithms like. Finally, an interface is developed for uploading, visualizing, and exporting PCB design layouts. The results showed that it can extract meaningful features from PCB layouts. While the CNN achieved high accuracy in detecting components and traces, challenges remain in handling multi-layered PCBs and overlapping components, which may require more advanced architectures. This experiential learning activity presents a novel approach to PCB layout redesign by combining Convolutional Neural Networks (CNNs) with optimization algorithms. With further integration into EDA tools and support for more complex designs, this approach can greatly streamline PCB development and be a powerful tool in modern electronic design automation.

Professional Development for Immersive Technologies: Assessing Teacher Educators' Readiness Before and After Workshop

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Abstract

Immersive technology, such as Virtual Reality (VR), are becoming increasingly widespread in our daily lives. Virtual reality has grown as a popular issue in recent years. This study explores the readiness of teacher educators for integrating immersive technologies, such as Artificial Intelligence (AI) and Virtual Reality (VR), in educational activities. A professional development workshop was planned and presented, integrating a pre-test and post-test assessment methodology to measure changes in preparedness levels. The research included a self-constructed questionnaire that measured awareness and usage, Perception and attitudes toward AI and VR, as well as the perceived benefits, obstacles, and future expectations for these technologies in education. Findings demonstrate a considerable improvement in educators' awareness and attitude with immersive technology after the training. The study highlights the relevance of professional development in bridging gaps in technology readiness, resolving issues such as cost, accessibility, and ethical considerations, and encouraging positive attitudes about incorporating AI and VR into teaching. These results underline the necessity of focused training programs to increase teacher educators' capacities, assuring their preparedness to leverage emerging technology effectively in the classroom.

Keywords: Immersive technologies, Artificial Intelligence, Virtual Reality and Professional development.

Recognition of Prior Learning and Its Impact on GER: A Study of Punjab's Higher Education Landscape

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Abstract

India's National Education Policy (NEP) 2020 sets an ambitious goal of increasing the Gross Enrolment Ratio (GER) in higher education to 50% by 2035. Despite a relatively high literacy rate, Punjab reported a GER of only 27.4% in 2021–22 below the national average highlighting the need for innovative and inclusive strategies to widen participation. This study explores the potential role of Recognition of Prior Learning (RPL) as a transformative mechanism for improving GER in the state, particularly in light of the UGC's National Policy on RPL in Higher Education (2025). The policy allows up to 30% credit transfer through RPL and is aligned with the Academic Bank of Credits (ABC) and the National Credit Framework (NCrF), offering a foundation for systemic change.

At present, formal pathways into higher education for experienced and skilled individuals, particularly those contributing to the economy through employment, innovation, or leadership are limited. RPL attempts to bridge this gap by formally recognizing experiential and informal learning and converting it into academic credits, thereby enhancing access, mobility, and equity in education.

A recent pilot implementation demonstrated the feasibility and value of RPL in facilitating academic progression. Building on this momentum, the UGC has initiated a phased rollout from the academic year 2025–26, aiming for system-wide adoption by academic session 2026-27.

This study examines the relevance of RPL in Punjab's socio-economic and educational context while drawing comparative insights from international models such as Australia's AQF (Australian Qualifications Framework) and South Africa's NQF (National Qualifications Framework), to understand its potential impact on GER and inclusive higher education.

Keywords: Recognition of Prior Learning (RPL), Gross Enrolment Ratio (GER), National Education Policy (NEP) 2020, Higher Education Access, Academic Credit Transfer

Learning the Learner: Redefining Knowledge Acquisition through Adaptive, Experience-Linked Learning Graphs

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Abstract

As learning heads into the era of personalization, traditional learning systems still fail to capture the richness of how people learn, relate, and apply knowledge. This paper advocates for an adaptive, experience-based knowledge graph architecture that represents both what is being learned and the individual's specific context in which it is taking place. Compared to conventional approaches that handle knowledge as immutable facts, this method constantly develops by merging information acquired from various sources—websites, books, videos, etc.—with the teaching approaches, examples, and conceptual connections that helped form the learner's comprehension. By adaptively mapping the depth, order, and style of every learning experience, the system generates a dense, contextual depiction of a learner's cognitive evolution. This tailored graph enables AI systems to give more instinctive feedback, navigate learners along personalized paths, and facilitate lifelong learning paths. The advocated architecture restructures knowledge acquisition beyond merely monitoring what is acquired but also how, when, and why—enabling a paradigm shift from fixed instruction to dynamic, learner-focused education.

Keywords: Personalized learning, knowledge graph, cognitive modelling, adaptive learning systems, context-aware education.

Bridging Theory and Practice in CAD through Engine Dissection and Experiential Learning

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Abstract

This research paper explores the inclusion of an experiential learning-based project in a second-year undergraduate engineering course, Computer Aided Design and Analysis, with the aim of enhancing student understanding by enabling learning from real mechanical systems, such as engines and gearboxes. To bridge the gap between theoretical concepts and practical application, a project based on experiential activity on internal combustion engine activity was introduced. In this hands-on project, students physically disassembled parts of the engine, created CAD models of its components and assembly, and performed mechanism design analysis. It enhanced their understanding of functional relationships, material properties, and the underlying design intent of the engine. Pre- and post-activity feedback from students yielded meaningful insights into the activity's effect on their engagement levels and conceptual comprehension. Supporting findings by other researchers also show that experiential learning improves student understanding and engagement with concepts through open-ended projects. This activity not only helped students apply CAD tools in a meaningful context but also highlighted the pedagogical value of incorporating real-world machines like an engine to strengthen both conceptual grasp and practical engineering skills. These advantages are empirically supported by a demonstrable improvement in course outcome scores for students engaged in experiential learning.

Keywords: Experiential Learning, Computer Aided Design (CAD), Engineering Education, Project-based Learning, Mechanism Design

Enhancing Education through Experience: A Study on Experiential Learning

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Abstract

From past few years, the interest among educators and researchers in experiential learning and its effects on student engagement and educational outcomes has been increased rapidly. This approach, which proclaims hands-on experience and reflection, has been recognised for making education more dynamic and impactful. These studies assess the significance and impact of experiential learning on students' education. The concept of experiential learning is widely discussed across various educational levels and disciplines. This article aims to investigate the meaning, nature, and effects of experiential learning in relation to various influencing factors on students' life. Studies consistently show that participants in experiential learning demonstrate higher engagement, improved critical thinking, content retention, confidence, and problem-solving skills compared to traditional instruction formats. These studies assess the significance and impact of experiential learning across educational levels and disciplines, including technology-enhanced, service, and project-based formats. This article investigates the meaning, nature, and effects of experiential learning including factors like emotional engagement, institutional support, and scalability on student motivation, academic performance, well-being, and life skills. Building on foundational work in the field most notably Dewey's principles of continuity and interaction and Kolb's seminal framework laid out in Experiential Learning work which systematically analyses a range of instructional approaches, case studies and content analyses. It explores how experiential learning enhances depth of comprehension, stimulates critical thinking, and promotes student-centred pedagogy.

Keywords: Experiential Learning, Student Education, Critical Thinking, Student-centred learning

PCB Extraction Using CNN and Deep Learning

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Abstract

Printed Circuit Boards (PCBs) are at the core of every electronic device, yet many older or reverse-engineered boards lack their original schematics. Without these diagrams, repair and analysis become slow and often inaccurate. In this work, we present an automated approach to bridge that gap by converting PCB images into circuit schematics. Our method uses image preprocessing to clean and segment board images, followed by feature extraction to detect traces, pads, and vias. These features are then represented as a connectivity graph, from which a netlist is generated and mapped onto standard schematic symbols. Experiments on single-layer PCBs show that the approach can reliably reconstruct connections and produce usable schematics. This reduces manual effort, speeds up reverse engineering, and opens new directions for applying machine learning to complex, multi-layer boards in the future.

Keywords: PCB reverse engineering, deep learning, YOLOv8, component detection, schematic generation, computer vision, layout extraction

Driven by Desire: Materialism, Curiosity, and Impulse Buying in a Gendered Context

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Abstract

Impulse buying has emerged as a prevalent behavioural tendency in contemporary consumer culture, often shaped by underlying psychological traits. This study explores the influence of curiosity and materialism on impulsive purchasing behavior, with an emphasis on gender-based differences. A good number of adults, aged between 18 and 25 years, participated in the study. Standardized instruments—the Curiosity and Exploration Inventory-II (CE-II), the Material Values Scale (MVS), and the Buying Impulsiveness Scale (BIS)—were administered to assess the respective psychological constructs. Data were analysed using Pearson's correlation, multiple regression, and independent samples t-tests. Findings indicated a moderate positive correlation between materialism and impulse buying ($r = 0.37$, $p < 0.01$), identifying materialism as a significant predictor of impulsive purchasing behavior.

Keywords: Curiosity, Materialism, Impulse Buying, Gender Differences, Consumer Psychology

How Experience Shapes Potential Entrepreneurs: A Systematic Review Validated by Neural Networks

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Abstract

Purpose: This study uses a systematic literature review (SLR) to investigate the complex relationship between experiential learning (EL) and many aspects of entrepreneurial success (ES), such as intentions (EIs), behaviour, and self-efficacy. Understanding how EL promotes entrepreneurial development is becoming more important as entrepreneurship education improves.

Methodology: A thorough search was performed in the Scopus database using selected keywords and Boolean logic to get peer-reviewed publications published in English. The inclusion criteria centred on empirical studies that examined EL and its impact on entrepreneurial constructs. A total of 65 articles were located and examined. To increase the validity of the theme findings, an Artificial Neural Network (ANN) model was used to validate important patterns and thematic clusters identified in the literature.

Results and Findings: The review demonstrates that experiential learning has a considerable impact on changing entrepreneurial intents and behaviour, increasing entrepreneurial self-efficacy and, in certain situations, predicting entrepreneurial success. Four major themes emerged: (1) EL in entrepreneurship education, (2) its impact on intention creation, (3) behavioural transformation through applied learning, and (4) the development of entrepreneurial self-belief. ANN validation showed the importance of EL's dimensions in formations of EIs of the young potential entrepreneurs.

Originality: This is one of the first studies to use SLR and ANN approaches to investigate the role of experiential learning in entrepreneurship. It presents a data-driven, validated review of the literature, laying the groundwork for future empirical and theoretical advances.

Keywords: Experiential learning (EL); Entrepreneurial intentions (EIs); Entrepreneurial self-efficacy (ESE); Artificial Neural Network (ANN)

Changing Paradigm in Higher Education Scenario

Bridging Ancient Wisdom and Modern Learning: The Indian Knowledge System in the 21st Century

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Abstract

The Indian Knowledge System (IKS) is an enormous collection of ancient wisdom that encompasses a wide range of fields, including philosophy, science, mathematics, medicine, the arts, and ecology, and is deeply rooted in India's cultural and intellectual heritage. Despite its significant contributions to global knowledge, IKS has been largely marginalized within India's mainstream educational framework, particularly since the colonial era. With the implementation of the National Education Policy (NEP) 2020, there is a renewed emphasis on revitalizing and integrating IKS into the formal education system in order to promote a more holistic, value-based, and culturally enriched learning environment. This study examines the many facets of IKS in the twenty-first century, emphasizing its applicability, difficulties, and possibilities for contemporary education. The study investigates important elements of IKS, including Ayurveda, Yoga, Vedic Mathematics, Astronomy, Environmental Ethics, and Indigenous Arts, and evaluates their suitability for use in modern curricula through a thorough literature review and policy document analysis. The alignment of IKS with international educational priorities like sustainability, mental health, and ethical consciousness is also covered in the paper. According to the study, there are several obstacles to incorporating IKS, such as a lack of genuine educational resources, inadequate teacher preparation, and prevailing attitudes that frequently minimize indigenous knowledge. It also highlights IKS's huge potential to support innovation, global identity, building a nation, and educational equity. The paper offers useful suggestions for legislative bodies, curriculum designers, and educators to guarantee meaningful implementation, drawing on successful case studies of institutions that have already incorporated IKS. The paper concludes that a balanced approach that combines ancient wisdom with modern pedagogical practices can enrich the educational landscape, promote cultural pride, and prepare students for the complex challenges of the future, all in line with NEP 2020's vision.

Keywords: Indian Knowledge System (IKS), National Education Policy 2020, Indigenous Knowledge Integration, Holistic Education and Educational Transformation

The Relationship between Fear of Negative Evaluation, Grit, and Academic Motivation among Detained College Students

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Abstract

In Traditional disciplinary practices in academic settings, such as detention it is often the aim to enforce compliance but may unintentionally suppress students' intrinsic motivation and long-term engagement. There is a growing recognition of the need for more reformist, task-oriented approaches that support psychological well-being and academic growth. This study examines the impact of Fear of Negative Evaluation (FNE) on two key constructs: academic motivation and grit. A sample of 100 adolescent college students participated in a survey. The results revealed a negative correlation between 'FNE' and intrinsic motivation to experience stimulation, and a positive correlation with a motivation and consistency of interests, a dimension of grit. No significant relationship was found between FNE and perseverance of effort. Regression analysis confirmed that FNE significantly predicts changes in both academic motivation and grit-related traits. These findings suggest that punitive, fear-driven systems may hinder meaningful learning and promote anxiety-driven goal fixation. The study highlights the need for educational environments to reduce reliance on punitive measures like detention and instead adopt supportive, autonomy-enhancing strategies.

Keywords: fear of negative evaluation, academic motivation, perseverance, consistency of interests

A Proposed Framework to Integrate Indian Knowledge System and Artificial Intelligence for Autism Spectrum Disorder

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Abstract

The population of individuals with Autism or Autism Spectrum Disorder (ASD) is steadily increasing. The Ancient Indian Knowledge System (IKS) is one of the oldest education systems in the world, focusing on the holistic development and integrated thinking of individual learners. Here, we propose a teaching methodology framework primarily designed for ASD learners. IKS and AI are integrated into this framework, which is tailored for individual learners, especially since ASD learners often prefer solitude. With the help of AI, learners are categorized based on their engagement levels and mapped with the ABC learning model to determine how lessons should be structured, encouraging continuous engagement. The framework also highlights the importance of yoga and meditation, which can benefit ASD learners by boosting concentration and mental peace. It promotes an inclusive educational environment by blending modern technology with ancient culture, creating an e-learning system suited for Education 4.0.

Keywords: Indian Knowledge System; Artificial Intelligence; Autism Spectrum Disorder; Teaching Methodology; Neuroplasticity;

Analysis on the Integration of Advanced Educational Methods through Generative AI to Enhance Teaching and Learning Process.

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Abstract

The integration of Generative Artificial Intelligence (GenAI) into educational frameworks represents a transformative shift in teaching, learning, and research methodologies. This paper explores the strategic incorporation of GenAI tools to enhance pedagogical practices, foster personalized learning experiences, and streamline academic research processes. By examining current applications and theoretical underpinnings, we identify key areas where GenAI can be leveraged to improve educational outcomes. The main goals of this analysis is to encompass the use of GenAI to customize educational materials according to the unique needs, preferences, and learning speeds of each student, thus promoting a more engaging and effective learning atmosphere that enhances Personalized Learning Experiences. Additionally, it involves the application of GenAI tools to streamline administrative responsibilities, including grading and content generation, which enables educators to focus more on interactive and student-centered teaching methods.

Keywords: Generative AI, higher education, personalized learning, research productivity, ethical considerations, strategic implementation

The Intersection of Sustainable Finance and Lifelong Learning: Performance and Policy Insights from India's ESG Thematic Mutual Funds

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Abstract

This study explores the intersection of sustainable finance and lifelong learning in the Indian context, examining how Environmental, Social, and Governance (ESG) thematic mutual funds perform and how educational frameworks can support India's carbon neutrality goals. By analysing the performance of eight ESG mutual funds from 2019 to 2024 and assessing curriculum reforms under the National Education Policy (NEP) 2020 and Outcome-Based Education (OBE), the research reveals a dual opportunity: ESG investments are financially competitive, and integrating sustainable finance into undergraduate and postgraduate education can enhance climate literacy and responsible investing behaviour. The study highlights that while ESG funds, such as Quant and ICICI Prudential, exhibit superior risk-adjusted returns, a significant gap persists in integrating ESG concepts into core academic content and lifelong learning initiatives. To close this gap, it advocates for a policy-driven educational model that fosters interdisciplinary learning, critical thinking, and sustainability competencies. This alignment of financial innovation with educational transformation is crucial for building an informed, climate-conscious citizenry that can steer India toward sustainable development and carbon neutrality.

Keywords: ESG, Sustainable Investment, Carbon Neutrality, Thematic ESG Mutual Funds, Sustainable Finance Education, Lifelong Learning, Education Policy, India, Climate Finance, Education System.

Needs of Teacher Education according to the Indian Knowledge System in India and Path Ahead with NEP- 2020

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Abstract

Knowledge of India will include Knowledge from ancient India. The pursuit of knowledge, wisdom and truth was always considered as higher human goals in Indian thought and philosophy. Indian Knowledge System based on Vedic literature, the Vedas and Upanishads. It comprises of Jnan, Vignan and Jeevan Darshan that have evolved out of experience, observations, experimentation and rigorous analysis. Indian culture and philosophy had a great influence on world. NEP 2020 recognizes this rich heritage of Indian knowledge and connect it to the modern era. These rich legacies to world heritage must not only be nurtured or preserved for posterity but also researched, enhanced and put to new uses through our education system. It aims to support and facilitate further research to solve contemporary societal issues. The core changes in the educational system must be centered on the teachers. The new education policy must make every effort to empower educators and support their efficient performance. In order for teacher education to be comprehensive, a teacher must process both traditional and current knowledge. Current IKS courses are linked to online learning environments. Modules for teacher orientation and training could be created to raise the standard of instruction in IKS course To train teachers on particular IKS-related topics, specialized teacher training facilities will be established. Grand National Challenges, National Competitions, Hackathons, and rewarding ideas will all be used to promote IKS innovations. By providing technological solutions that highlight Indian heritage to both Indians and the global community, IKS will advance heritage technology. With skill-based programs, it hopes to secure 10% of global tourism and offer our young people enormous job opportunities.

Keywords: - Indian Knowledge System, Teacher Education, skill based programs, Employment Opportunities, NEP 2020

Evolution of Education Policy in India: A Historical and Analytical Perspective

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Abstract

India's education system has changed dramatically over the years, evolving from colonial-era frameworks to a more inclusive and learner-friendly approach. This paper looks at how education policies in India have developed—starting from the British period, moving through post-independence reforms like the National Policies on Education in 1968 and 1986, and arriving at the transformative National Education Policy (NEP) 2020. These shifts reflect the country's efforts to make education more accessible, relevant, and responsive to modern needs. By exploring the key ideas behind these policies and how they were implemented, this study highlights what has worked, what challenges remain, and what can be improved. The paper offers a thoughtful look at how policy can shape not just systems, but the lives and futures of learners across India.

Keywords: Indian Education System, Colonial to Contemporary Education Reforms, Educational Transformation, Equity and Access in Education.

Bibliometric Analysis for the Development of Research Problem: A case study

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Abstract

In the past two decades, several studies have been reported on developing a research problem, especially for research gap analysis in the higher education system. Researchers have used several pedagogical measures to establish the validity of the research problem. However, little has been reported on using bibliometric analysis based on networking diagrams and occurrence tables for developing research problems in the process industry with constraints such as cost, environmental concern, government policies, regulatory compliance, etc. This study systematically reports the application of bibliometric analysis using the Web of Science and Scopus databases on the secondary recycling of polyamide-6 reinforced with bakelite for gun frames. The study outlines that the composite materials selected based upon the bibliometric analysis offer a promising solution requiring recycling polymer waste and demonstrate the potential of combining thermosets with thermoplastic polymer to create high-performing materials and reduce the harmful effects by recycling. The study also highlighted the need to develop a functional prototype per selected composition for 3D printing of tensile, flexural, and compressive coupons as per ASTM, optimise the process, and investigate prepared functional prototypes' thermal, mechanical, dielectric, and morphological properties. The results suggest that using bibliometric analysis, the objectives were well defined as required for the process industry, which complies with most of the requirements of government policies, regulatory agencies, environmental concerns, etc., and provides a promising solution for recycling plastic waste.

Keywords: Bibliometric analysis; Research gap analysis; Research validity; Recycling; Thermoplastics

Importance of Institutional Offline Library in Hybrid Educational Pedagogy: An Indian Perspective

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Abstract

When the COVID-19 pandemic hit the world, the entire educational pedagogy has changed to a hybrid model. The kids were able to continue their regular education despite the unparliamentary situation thanks to the widespread availability of digital content and easy access to the internet. The school, colleges, and universities have closed their offline libraries and encouraged the students to use digital resources. Educationists are now concerned about the applicability of offline libraries in the hybrid education pedagogy in the new normal mode. The current study examined the footfall in an Indian institutional library before and after the COVID-19 pandemic in an effort to manipulate its relevance. All parties involved would be benefited from the analysis in understanding the offline libraries' altered role in the traditional educational setting, especially in a developing country.

Keywords: COVID-19, Hybrid Educational Pedagogy, Library Footfall, Offline Library, Online education

Ensuring Access, Equity, and Inclusivity: The Role of Translanguaging in NEP 2020's Educational Framework

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Abstract

India's multilingual and multicultural landscape offers both opportunities as well as challenges for setting up an educational structure that meets the needs of various communities of society. Over the past few years, translanguaging has emerged as a powerful pedagogical tool that facilitates effortless shift between two or more languages in educational settings so as to improve students' understanding of concepts and boost their academic achievement. The current study examines translanguaging within the context of the National Education Policy (NEP) 2020, which prioritizes enhancing equity and inclusivity in the curriculum design, ensuring that students from diverse linguistic backgrounds have equal access to all the opportunities across all educational stages—from early schooling to higher education. Translanguaging, which allows bilinguals and multilinguals to move fluidly between various languages, has become a key approach for materializing this goal, and fostering enhanced comprehension, cognitive process, and academic success. By an analysis of pre-existing data, this study explores how translanguaging practices transcend conventional monolingual teaching methods and cultivate a more active and learner-centered educational environment. By aligning translanguaging with NEP 2020's emphasis on comprehensive and holistic education, the paper seeks to uncover its key role in facilitating equity, accessibility, and inclusivity in higher education. The findings of the research illuminate the effectiveness of translanguaging as a pedagogical method that serves a crucial role in both expanding students' understanding and developing sharp analytical abilities, ultimately bolstering academic integration in multilingual educational environment. In conclusion, the paper advocates for a fluid and flexible framework designed to be inclusive of linguistically diverse communities, cultivating a teaching-learning environment where linguistic diversity is recognized as an asset, not a hindrance. Ultimately, the research contributes to the ongoing discourse on multilingual education, highlighting the significance of translanguaging as a transformative pedagogy for students seeking academic achievement and navigating cultural and linguistic landscape.

Keywords: NEP 2020, Translanguaging, Academic Achievement, Higher Education, Inclusivity

Assessment of NEP 2020 Implementation Using Force Field Analysis: A Systems Perspective on Policy Enablers and Resistors

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Abstract

The National Education Policy (NEP) 2020 presents a transformative framework for reimagining India's education system in alignment with 21st-century needs. This study applies Lewin's Force Field Analysis (FFA) to assess the practical implementation of NEP principles by identifying the key driving and restraining forces influencing policy enactment. The research categorizes NEP's foundational principles into core themes and evaluates implementation enablers and barriers through qualitative insights from policy literature, institutional practices, and expert perspectives. The analysis reveals significant momentum in areas such as digital learning and curriculum restructuring, while challenges persist in teacher readiness, infrastructural gaps, and systemic inertia. The paper concludes with recommendations to strengthen implementation by leveraging driving forces and minimizing resistance, thus enhancing the realization of NEP's vision of equitable and holistic education for all.

India's National Education Policy (NEP) 2020 is a landmark reform aimed at overhauling the education system to make it more inclusive, flexible, multidisciplinary, and aligned with global standards. Rooted in Indian ethos, the policy seeks to ensure universal access to quality education and prepare learners for the challenges of a rapidly evolving global landscape. NEP 2020 emphasizes student-centric learning, foundational literacy, teacher empowerment, technology integration, and equity in education delivery. Despite the clear vision and policy direction, implementing NEP 2020 presents significant challenges. Translating high-level principles into practice involves navigating institutional inertia, capacity gaps, and stakeholder resistance. Therefore, assessing the forces that support or hinder its implementation is essential.

This paper uses **Force Field Analysis (FFA)** to evaluate the dynamic interplay of driving and restraining forces influencing the implementation of NEP 2020. By identifying these forces, this research contributes actionable insights for policymakers, educators, and institutions striving to actualize the NEP's transformative goals. Force Field Analysis is a diagnostic tool used to evaluate the factors that support or hinder change. Developed by Kurt Lewin, it conceptualizes change as a balance between **driving forces** (which push for change) and **restraining forces** (which resist change). The net outcome of these opposing forces determines the extent and effectiveness of change implementation. In this study, FFA is applied to grouped themes derived from NEP's foundational principles. Each theme is assessed for **Driving Forces** (policy enablers, supportive initiatives, stakeholder momentum) and **Restraining Forces** (institutional, cultural, infrastructural, or regulatory barriers). The sources of information and data used for the study include; Policy documents and official NEP 2020 guidelines, academic literature and implementation reports and expert opinions from educators and administrators

Artificial Intelligence (AI) in Education

Assessing AI Chatbots in Engineering Education: Performance Analysis for GATE 2024 in Civil and Computer Science Engineering

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Abstract

The increasing adoption of AI chatbots in engineering education necessitates a thorough valuation of their effectiveness for national-level competitive examinations in India. This study assesses the performance of advanced chatbots, specifically Grok and Perplexity, in solving the Graduate Aptitude Test in Engineering (GATE) 2024 question papers for Civil Engineering (CE) and Computer Science and Engineering (CSE) streams. The GATE exam, a rigorous national-level assessment, evaluates the technical proficiency of engineering graduates. We analyzed the chatbots' performance based on accuracy, explanation quality (clarity, completeness, and correctness), and error types (conceptual, calculation, reasoning, and misinterpretation). The study covered multiple-choice questions (single and multiple correct answers) and fill-in-the-blank questions, categorized by format (text, numerical, image-based, and code-based). Both chatbots achieved qualifying GATE scores ranging from 63 to 73 out of 100 for CE and CSE, though their scores were lower than those of All India Rank #1 holders. Grok outperformed Perplexity in the CSE stream, while Perplexity scored higher in the CE stream. The chatbots demonstrated higher accuracy on text and numerical questions compared to image-based ones, with misinterpretation of images and data being the most frequent error. Further analysis explored chatbots performance across sub-areas within CE and CSE. This study highlights the strengths and limitations of each chatbot, offering valuable insights into their capabilities for solving complex, domain-specific problems and their potential as educational tools for students preparing for national-level competitive examinations.

Keywords: AI chatbots; Grok; Perplexity; GATE; engineering education

Role of Artificial Intelligence in Shaping Cognitive Skill Learning

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Abstract

Artificial Intelligence (AI) is promptly transforming the landscape of educational industry by providing a wide range of techniques to develop and optimize the cognitive skills among the learners. The various cognitive skills such as attention, memory, logical reasoning, problem-solving, and critical thinking are fundamental to academic success and lifelong learning. AI technologies offer dynamic, adaptive, and personalized learning experiences that can be tailored to meet each learner's unique cognitive profile.

This paper explores the multifaceted role of AI in shaping cognitive skill learning through intelligent tutoring systems, adaptive learning platforms, educational chatbots, and data-driven feedback mechanisms. These AI-driven tools not only track and analyze learner behavior but also provide real-time feedback, enabling learners to reflect on their thought processes, monitor their performance, and adjust their strategies accordingly. With the use of natural language processing and machine learning, AI can accurately interpret the inputs of the students, can correctly assess their level of learning and understanding and helps to reshape the cognitive learning among them.

Artificial Intelligence also facilitates the immersive learning experiences by the use of advanced technologies such as augmented reality, virtual reality which permits students and learners to get engaged in simulating the environment that involves reasoning, decision making and helps them to solve the complex real-world problems. These types of tools are effective in the field of special education, STEM education where traditional methods of learning fail in justifying the cognitive needs of the learners. To conclude, it can be stated that Artificial Intelligence significantly plays and important role in shaping the cognitive skills and learning through enabling of interactive, scalable and personalized solutions of learning. It also supports the learners and educators in AI can support educators in nurturing the people in cognitive learning and helps them to shape their skills in tackling the complexities of 21st century.

Keywords: Artificial Intelligence, Cognitive learning, Skill learning, Virtual reality, Augmented Reality, Education

Integration of Artificial Intelligence in Flipped Method of Teaching: A Mini Review

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Abstract

In higher education, flipped method of teaching has gained widespread acceptance as an effective pedagogical technique. Being prepared in advance helps students to grasp the concepts taught during classroom sessions, which is also a crucial requirement of flipped classroom. With this teaching approach, teachers involve students in a variety of discussions, experiments, and activities throughout the class. The introduction of technology has completely changed how students learn, making it simpler, more interesting and approachable, more than in the past. These days, students learn at home by viewing videos on the internet, reading the accessible materials, and participating in different activities during classroom sessions. Artificial intelligence (AI) is one such technology which has the capacity to seamlessly deploy flipped teaching style. AI facilitates learning and has the potential to greatly enhance and revitalize by delivering affordances that support the student's individualized learning experiences. This mini review explores the various models of flipped method of teaching, integration of AI in flipped classrooms, and also highlighting its potential benefits like improved learning outcomes and scalability. However, it also aim to provide an overview of the opportunities and challenges that AI presents in the context of flipped method of teaching.

Keywords: Flipped model, Teaching, AI in education, Integration of AI, Outcomes

Role of Blended Learning and Artificial Intelligence in Student Engagement for the Learning of Concepts of Biotechnology

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Abstract

The quick development of Artificial Intelligence (AI) has impacted the education system immensely, providing new ways of learning and instruction in applied sciences like biotechnology. The study examines the application of AI to develop biotechnology education in terms of three central aspects: understanding concepts, efficiency in learning, and comparative effectiveness in teaching. First, AI-based tools like intelligent tutoring systems, virtual labs, and interactive simulations have been effective in making complex biotechnological processes simple and easy to visualize, thus allowing students to understand basic concepts like gene editing, fermentation, and metabolic pathways more clearly. Secondly, AI-facilitated blended learning environments, where traditional teaching is supplemented with customized AI interventions, have shown a significant decrease in learning time through real-time adaptive content tailored to individual learning preferences and automatic real-time feedback. Lastly, comparing AI-assisted instruction with traditional teaching strategies indicates that the use of AI tools directly improves student engagement, conceptual retention, and assessment scores. Based on evidence from recent IEEE and Scopus-indexed literature, this article brings to the fore the potential of AI to revolutionize contemporary biotechnology education and urges the intentional adoption of its application in academic curricula.

Keywords: Artificial Intelligence, Blended Learning, Biotechnology Education, Student Engagement, Conceptual Learning

The AI-Literate Educator: A Technical Framework for Faculty Development in Outcome-Oriented Learning Using Artificial Intelligence

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Abstract

The growing incorporation of artificial intelligence in higher education offers novel opportunities for customizing instruction, refining evaluation, and augmenting learner outcomes. The successful implementation of these technologies relies on the preparedness and proficiency of educators. This research introduces a systematic, technically robust approach for cultivating faculty members skilled in the implementation of artificial intelligence in outcome-focused teaching and learning settings. The proposed paradigm is grounded in modern educational theory and aims to assist educators through successive developmental stages, from basic comprehension to leadership in technology-enhanced teaching methodologies. Utilizing a publicly accessible educational dataset, we model faculty growth trajectories and employ predictive modelling approaches, such as logistic regression and recurrent neural networks, to demonstrate how data can enhance decision-making in professional learning environments. An analysis of feature contributions aids in identifying the components of training that most significantly influence educator advancement. The paradigm prioritizes alignment with instructional objectives, reflective teaching methodologies, and quantifiable learning results over merely technological competencies. It provides educational institutions with a scalable method for enhancing capacity in artificial intelligence-driven instruction while preserving pedagogical integrity. This study presents a data-driven approach to faculty development that fosters instructional creativity and sustains educational excellence across time.

Keywords: Artificial intelligence, faculty development, educational technology, outcome-based education, pedagogical innovation, AI literacy

Transformative Effect of AI in Education

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Abstract

Artificial Intelligence (AI) is revolutionizing the education sector by fundamentally reshaping how learning is delivered, assessed, and managed. As AI technologies evolve, they offer unprecedented opportunities to personalize education, automate administrative tasks, and improve access to learning resources globally. This research paper delves deeply into the transformative effect of AI in education by exploring the historical evolution, key applications, real-world implementations, and the socio-ethical implications of AI-driven education. The study also highlights emerging trends, data-driven insights, and strategic recommendations for adopting AI responsibly in diverse educational settings. The paper aims to present a comprehensive, evidence-based outlook on how AI is empowering educators, engaging learners, and enriching institutional decision-making.

Keywords: Artificial Intelligence, Education Technology, Personalized Learning, Intelligent Tutoring Systems, EdTech, AI Ethics, Automated Assessment, Educational Administration

Deep Learning Framework for EEG-Based Mind Wandering Detection in Education

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Abstract

Mind wandering (MW) is a widespread phenomenon in educational settings where the learner's attention involuntarily drifts away from the primary learning task towards task-unrelated thoughts, thus affecting learning outcomes. The detection of MW must be accurate so as to build adaptive and personalized learning technologies. In this study, conducted in the context of a learning setting, we present a deep learning-based framework for MW detection using electroencephalography (EEG) data. For practical reasons, EEG data collection in the real-world classroom setting becomes inconvenient. Hence, a publicly available EEG dataset originally recorded in a meditation setting was repurposed to mimic MW and focused attentional states. The EEG signals were pre-processed in EEGLAB, with several transformation steps producing two-dimensional scalp maps depicting spectral power across standard frequency bands. For classification of attentional states, a hybrid deep learning architecture combining convolutional neural network (CNN) layers for spatial feature extraction and long short-term memory (LSTM) layers for temporal sequence modelling was then adopted. Experiments are underway, and the results will be reported in the full paper with proper comparison with existing methods.

Keywords: Mind wandering, Personalized learning, Deep Learning, EEG

Educate Elevate Innovate: Integrating Generative Artificial Intelligence and Neuromarketing to Enhance Pedagogical Effectiveness

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Abstract

In this hyper-connected world, the attention of learners is fragmented with the increase in distractions at their disposal. These distractions are increasing the diversity of learners by shaping how they process information, form habits, and engage with learning environments differently. In the meantime, traditional teaching methods with their one-size-fits-all approach and linear instruction are no longer adequate to meet such varied needs of the learners. To address these concerns, our conceptual paper aims to explore the integration of new age technologies like Generative Artificial Intelligence (GenAI) and neuromarketing to improve pedagogical effectiveness. With the advent of GenAI, it is now possible to develop customized content at a very large scale. Meanwhile, neuromarketing helps in comprehending how people react to different messages, visuals and content. This understanding could include identifying what grabs attention, what sticks in the memory, what could trigger an emotional response. Educational content designed with neuromarketing's insights and tailored capability of GenAI could reach more people and could make a stronger impact in the lives of the learners. This impact can be in the form of better engagement, improved learning outcomes and meaningful student-educator discussions. For doing so, our study synthesizes both domains and proposes Neuro-GenAI pedagogical framework to enable personalized learning experiences that are both emotionally resonant and cognitively efficient. This framework will provide a foundation for empirical testing and classroom innovation in the age of AI. Practically, it will empower curriculum innovators and pedagogical strategists worldwide to develop adaptive learning environments.

Keywords: Generative Artificial Intelligence, Neuromarketing, Pedagogical Effectiveness, Personalized Learning, Neuro-GenAI Pedagogical Framework

Factors Impacting Ethical Usage of Artificial Intelligence in Higher Education Institutions: A Systematic Literature Review Approach

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Abstract

The Prompt growth of generative AI technologies is revamping the education sector by enabling personalized learning, automating tasks, and enhancing academic efficiency. Such advancement can greatly boost the quality of teaching and learning outcome. However, with the advantages the usage of AI also bring number of critical concerns regarding the misuse and unethical usage of AI in the academia thereby increasing the risk of plagiarism and academic dishonesty in academic related work carried out by both students and faculty members alike.

While dealing with this misuse and risk pertaining to AI in academia it is essential to ensure development and formulation of a clear, ethically just and actionable AI policy in each higher educational institutes for ensuring the integrity and advancement of the complete educational system.

The purpose of this presented study is to explore various factors that impact sound AI policy formulation in Indian higher education institutions which is becoming both quite essential and critical at same time. Preliminary review of literature reveals leadership and strategic visioning, Stakeholder involvement, Governing and legal framework along with ethical consideration as few of the important factors impacting the AI policy formulation along with its implementation in Indian higher education system. This study further aims to develop a literature backed comprehensive conceptual model for the aforementioned purpose by carrying out an extant review of literature. The study also intent to examine the reasons and rationale behind the slow adoption and formulation of AI related policies by the Indian higher education institutes along with the some of the ways to overcome the same.

Keywords: Artificial intelligence, AI Policy Formulation, Higher Education, Ethical usage, AI integration

A Review of Literature on Factors Influencing the Adoption of AI Tools in Higher Education

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Abstract

Modern higher education is undergoing a significant reorientation, as rapid advances in the field of artificial intelligence, in particular generic models such as ChatGPT, are systematically integrated into pedagogy, evaluation, research practice and institutional governance. Initially, these advanced systems, capable of producing text, images and other media similar to people, have rapidly evolved into practical instruments adopted within the training, administrative and research functions of higher education institutions. The rapid spread of these instruments has led to a profound paradigm shift, encouraging educational institutions to rethink traditional teaching methods and adopt new pedagogical strategies.

(Ahmed Tlili et al., 2023) described ChatGPT as a central case study, highlighting how the collaboration between people and machines develops within educational contexts. (Wach et al., 2023) reinforces this perspective, pointing out that the potential negative effects of generative AI are clearly being converted to different sectors, especially within education. (Mogavi et al., 2024) provide additional evidence of this rapid recording, using early adopters accounts to alleviate patterns of AI use and attitudes in different educational contexts, with particular emphasis on higher education.

The evidence of the rapid involvement of the academic community with these technologies is evident in the wave marked by the surge in publications related to ChatGPT in 2023, which Tikhonova and Raitskaya (2023) interpret as indicating a model of rapid incorporation into academic speech and practice. At the same time, Birenbaum (2023) describes the growth of chatbots as a chatbot marked for education, debating whether these systems constitute constructive disruptive agents or present truly destructive threats to established educational practices. Currently, literature has largely focused within. The swift tempo of technological change frequently outpaces institutions' ability to develop corresponding policies, pedagogical frameworks, and ethical guidelines. With the fluid nature of the adoption context, the sector continues to shift, requiring organizations to rely on temporary, reactive responses that often come before comprehensive future-oriented planning strategies. As a result, research on AI adoption is still relatively initial in nature, as it primarily identifies early perceptions and immediate impacts rather than long-term impacts or full-fledged integration strategies. Given the evolving nature of the technology and its use, a long-range, adaptive research agenda is required.

Role of Artificial Intelligence and Machine Learning in Teaching and Learning

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Abstract

The recent advances in Artificial Intelligence (AI) and Machine Learning (ML) are significantly transforming the field of education. Digital Transformation and Educational Technologies are improving teaching methodologies and learning outcomes, particularly in the context of innovative and evolving subject areas. AI-powered tools such as intelligent tutoring systems, adaptive learning platforms, and automated assessment engines enable personalized, flexible, and data-driven instruction. Machine learning algorithms help identify learning patterns, predict student performance, and provide real-time feedback, allowing educators to customize teaching strategies for diverse learners. Additionally, AI supports content creation, curriculum updates, and immersive learning environments through virtual labs and simulations. While these technologies hold immense promise for revolutionizing education, the paper also addresses challenges related to ethical concerns, data privacy, and the digital divide. By examining current applications and future trends, this research highlights how AI and ML can foster a more inclusive, engaging, and innovative learning ecosystem.

Keywords: Artificial Intelligence, Machine Learning, Digital Transformation, Learning outcomes, Education Technology.

AI-Driven Transformation: Revolutionizing Research, Teaching, and Administration in Higher Education

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Abstract

Academic research, teaching, and administration are the three key areas of higher education that artificial intelligence (AI) is transforming. The rapid advancement of AIML is reshaping higher education the study examines the role of AI in various aspect. AI-driven tools in academic research facilitate practical literature synthesis, predictive analytics, and complex data interpretations

AI in academic teaching AI-powered tools such as intelligent tutoring facility automatic grading and adaptive learning platforms students' engagement for better learning and also uses curriculum planning for future-ready competencies, improves cognitive skill development through immersive simulations, and personalizes learning through these platforms.

In that area some challenges are there like data security and its privacy is a major concern about it and this are also obstacle to use AI resources this type of issue must be addressed by institute level & also academic administration play a important role in terms of working efficiency major contribution of AI help in resource allocation easy of admission & enrolment process &also get a quality work and actual time feedback system

This study investigates how and where AI is transforming Education 4.0 and beyond, how it may f aster efficiency while overcoming technical and ethical issues. It is in aline with THEC2025's subj ect of "Transforming Higher Education for Future."

Also ensuring AI integration supports sustainable education. Strong literacy initiatives for teachers and students, moral frameworks for data management, and expansion of infrastructure are some of the strategic concepts forutilizing AI's potential. The findings are under scope the need of collaborative effort between academia and industry they shape the future of AI-driven education.

Keywords: Artificial Intelligence, Academic Transformation, Education 4.0, Ethical Challenges, Adaptive Learning

Eye Luminate – Smart Eyes for the Blind in the Context of India to Learn

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Abstract

One of the most challenging health conditions worldwide is visual impairment or blindness which affects even simple daily activities like locomotion and information retrieval. The current research investigates and represents the proposed smart glasses as a wearable technology to address the challenge of learning among physically disabled people. This study assesses different algorithms such as YOLO v5, Tesseract OCR, gTTS- Google Text-to-Speech for instance, object detection, text recognition and to provide audio feedback to assist learning task and improve the quality of life of visually impaired people. The study also examines smart glasses' user-friendliness and comfort factors, with suggestions provided where appropriate. The proposed smart glasses potentially have a huge impact on enhancing the daily life experience of blind people with a more advanced method for perceiving the environment and receiving data to learn with the help of advancements in technology, and make themselves employable.

Keywords: Smart Glasses, Wearable Device for Visually Impaired People, Learning, Employment

A New Era of Education with the Integration of AI Tools

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Abstract

Artificial Intelligence (AI) is playing a significant role in simplifying many professional tasks. With the onset of its application in Education, it has significantly eased the burden with respect to a lot of activities associated with the teaching profession. It is aiding educators in multiple tasks such as content creation, lesson planning, assessments, etc.

However, many educators have still not taken the benefit of this AI revolution to its full capacity. In this work, a carefully curated questionnaire was designed to reach out to the educators and get their insights on how AI has changed their teaching methods, and if it has helped in improving efficiency and student learning outcomes.

It has been showcased how AI tools are making a positive impact in the field of education.

Various ways in which educators are integrating AI tools in their teaching methods are also presented. The work also highlights the specific tools that are being used by the educators and their awareness regarding the ethical implications associated with the integration of AI tools which is a significant challenge in AI tool usage.

Keywords: Assessment; Education; Ethics; Outcomes; Teaching

Advances in Educational Technology

Enhancing Learning Outcomes through a Student-Centred, Application-Based Approach in Virtual Instrumentation

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Abstract

This study explores the effectiveness of a student-centred, application-based teaching-learning approach in enhancing both conceptual understanding and practical implementation among undergraduate engineering students. The intervention was conducted as part of a final-year course titled *Virtual Instrumentation*, offered within the Electronics (Instrumentation and Control) Engineering program at Thapar Institute of Engineering and Technology. A cohort of 20 students participated in the study, which employed a case study methodology to assess the impact of the instructional strategy.

The approach emphasized active learning techniques such as "Think, Pair, and Share," enabling students to collaboratively engage in problem definition, conceptual application, and solution analysis. This method fostered deeper comprehension, encouraged peer-to-peer learning, and enhanced programming proficiency, especially among students with initially limited coding experience. The interactive nature of the approach not only made the subject matter more engaging but also contributed to the professional development of instructors through exposure to diverse student insights. Overall, the intervention demonstrated that student-centred, application-driven pedagogy can significantly improve learning outcomes and better prepare graduates for real-world engineering challenges.

Keywords: Virtual Instrumentation, Application-based Learning, peer-to-peer learning, application-driven pedagogy

Enhancing Student Engagement and Learning Through Augmented Reality: A Review

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Abstract

Augmented Reality (AR) overlays digital information onto the physical world, offering situated, interactive, and visually rich learning experiences. This review synthesizes recent evidence on how AR influences student engagement and learning across K–12 and higher education. We summarize theoretical underpinnings, key findings on engagement and achievement, moderating factors (e.g., subject domain, interaction modality, duration, assessment design), and design principles for effective AR-enhanced instruction. We also outline implementation models, accessibility and equity considerations, and future research directions. Overall, the literature indicates that well-designed AR can improve behavioural and cognitive engagement, conceptual understanding, spatial reasoning, and knowledge retention, with outcomes contingent on alignment to learning goals and careful management of cognitive load.

Keywords: Augmented Reality, student engagement, learning outcomes, cognitive load, motivation, instructional design, immersive learning

A Study on the Use of Animated Visualizations in Engineering Drawing Education

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Abstract

Engineering Drawing (ED) is a compulsory first-year course for all engineering disciplines of undergraduate (UG) curriculum at TIET Patiala. The course demands strong spatial visualization skills. However, traditional pedagogical approaches of presentation slides containing static diagrams can be difficult for students to interpret and visualize, especially for complex topics like ‘Projections’ and ‘Development of Surfaces’. Animated visualizations offer a potential solution to enhance learning by making abstract concepts more accessible. This study explored how animated visualizations improve student understanding in Engineering Drawing (UES-101) course at TIET Patiala during the Jan-May 2025 semester. Involving about 160 undergraduate students, the study used a pre-quiz to assess baseline knowledge, followed by animations created by the authors to illustrate spatial concepts on two topics: ‘Projections of Points’ and ‘Development of Surfaces’. A post-quiz measured the learning outcomes. Analysis using *t*-tests showed statistically significant improvements in the class average in both topics (p -value < 0.001). Analysis showed a notable increase in average scores from 2.9 to 3.5 (out of 5) for ‘Development of Surfaces’ and from 5.2 to 5.8 (out of 8) for ‘Projections of Points’, suggesting animations enhance conceptual clarity and spatial visualization, particularly for students struggling with static diagrams. Further analysis was performed using two-factor analysis of variance (ANOVA) to study the effect of gender and student groups on their pre-and post-quiz scores. The dynamic nature of animations helped students better grasp complex spatial relationships, making abstract ED concepts more tangible.

Keywords: engineering drawing; animation; visualization; engineering education

Faculty-Led Curriculum Innovation: A Consensus-Based Model for Syllabus Enhancement

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Abstract

In an era of rapidly evolving knowledge and industry demands, static academic syllabi often fall short in equipping students with current and relevant competencies. This research proposes a dynamic, collaborative approach to curriculum advancement through a multi-instructor, course-led enrichment model. In this method, each instructor independently explores and studies a contemporary or emerging topic not currently included in the official syllabus. These topics are then presented to fellow instructors in a structured format. Following all presentations, a group discussion is held to deliberate the relevance, feasibility, and impact of each proposed topic. A consensus or voting mechanism is employed to select the most valuable topics for inclusion in the curriculum. The selected content is then integrated into course delivery, ensuring students gain exposure to updated, interdisciplinary, and practical knowledge. This paper presents the consensus based curriculum enhancement model, its pedagogical benefits, and challenges, arguing that such a collaborative, faculty-driven process fosters continuous syllabus innovation and may significantly enhances student learning outcomes.

NSBT3A Model of LMS Integration in Educational Organisations: Streamlined Guidelines to Encourage Effective Learning

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Abstract

This paper presents the NSBT3A Model of Learning Management System (LMS) Integration in educational organisations. The model is designed to optimize the use of LMSs by aligning technical, administrative, and pedagogical needs. By outlining a series of steps—from organisational needs assessment to system adaptation and execution—the paper details how a strategic integration can enhance accessibility, streamline workflows, and promote personalized learning. The discussion also addresses benefits and challenges, providing a guideline for successful implementation.

Keywords: LMS Integration; Educational Organisations; NSBT3A Model; E-learning; Digital Education

Offline Digital Assessment System: Real-Time Quiz Administration and Evaluation through Local Server Networks

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Abstract

Traditional assessment methods in engineering education often face challenges related to immediate feedback, efficient evaluation, and security concerns associated with internet-dependent systems. The reliance on external networks for digital assessments creates vulnerabilities including connectivity issues, potential cheating through unauthorized internet access, and delays in result compilation. This research presents the development and implementation of an innovative offline digital assessment system that enables real-time quiz creation, administration, and evaluation within classroom environments through localized server networks, eliminating dependency on internet connectivity while maintaining digital assessment advantages.

The proposed system operates through a dedicated local WiFi network hosted on a classroom server, creating a controlled digital environment accessible only to students within the physical classroom space. Faculty members can create customized quizzes instantly using an intuitive interface, distributing assessments to student devices through the local network without requiring internet access. The system architecture ensures that student devices can only connect to the assessment platform while blocking external internet access, thereby maintaining academic integrity and preventing unauthorized resource consultation during evaluations.

Implementation across Mechanical, ECE, EEE, IT, and CSE departments demonstrated the system's versatility in handling diverse question formats including multiple-choice, descriptive answers, and technical diagrams. The platform supports both Tamil and English languages, accommodating the multilingual student population effectively. Real-time response tracking enables faculty to monitor quiz progress, identify completion status, and generate immediate analytics upon submission. The automated evaluation system provides instant feedback for objective questions while facilitating efficient review processes for subjective responses.

Results indicate significant improvements in assessment efficiency, with quiz creation time reduced by 60% and evaluation processes accelerated by 75% compared to traditional methods. Student engagement levels increased notably due to immediate feedback mechanisms and interactive digital interfaces. The system successfully eliminated common technical disruptions associated with internet-dependent platforms while maintaining robust data recording and analysis capabilities. Faculty reported enhanced ability to conduct frequent formative assessments, enabling better understanding of student comprehension after each topic. This research contributes to the advancement of educational technology by demonstrating how localized digital solutions can address practical classroom assessment challenges while maintaining academic standards and technological advantages.

Keywords: Offline Assessment System, Local Server Network, Real-Time Evaluation, Digital Quiz Platform, Classroom Technology

A Journey of Paradigm Shift for E-Content Development during Pandemic and Post Pandemic Times – A Case Study of Effectiveness of E Learning

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Abstract

The COVID-19 pandemic compels higher education to rapidly adopt digital teaching models, redefining how learning is structured and delivered. This paper corresponds with the principles of Education 4.0 and contributes to the ongoing shift in engineering education by showing how structured use of simulation, multimedia, and digital tools can sustain effective learning under restrictive conditions. This transition is illustrated through the digital reformation of the Robotics Engineering course at Thapar Institute of Engineering & Technology (TIET), aligning closely with the core principles of Education 4.0. The course is restructured using Outcome-Based Education (OBE) supported by a Learning Management System (LMS) that manages lectures, assessments, and communication, with integrated tools for structured quizzes and transparent evaluation.

A key component of this transition is the use of simulation software RoboAnalyzer that replaces in-person lab sessions and enables students to visualize kinematic and dynamic concepts through virtual robotic models enhancing conceptual understanding and problem-solving skills. To enhance visual learning, a complete set of video lectures is developed using Camtasia, featuring animations and diagrams to simplify complex robotics concepts. These lectures are published on a dedicated YouTube channel where they have received over 19,000 views, reflecting improved reach and their contribution toward transforming higher education for the future.

Keywords: Outcome Based Education (OBE), Learning Management System (LMS), Robot Kinematics and Dynamics

Advances in Educational Technology: Adaptation to New Technologies

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Abstract

Technology has drastically changed our lives, altering many aspects and reshaping daily encounters. Modern advances have made it possible to carry out complex and critical tasks with ease. Education is one of the major domains that technology has revolutionized. Quality education is a critical component of the United Nations' Sustainable Development Goals 2030 Agenda, which aims to ensure inclusive and equitable education for all. Digital tools have emerged as critical assets for achieving this goal. The COVID-19 epidemic has highlighted the importance of digital technology in education, resulting in a significant shift in the system.

Digital technologies are increasingly used as knowledge producers, co-creators of information, mentors, and evaluators. Technological innovations have made students' lives easier by replacing old pen and paper with software and tools for presenting and generating projects. Digital gadgets such as iPads and e-books provide lightweight and convenient alternatives to bulky notebooks and textbooks, boosting interest in study. This study examines the importance of digital technologies in education, focusing on their key uses and educational obstacles.

Quality education is a key component of the United Nations' 2030 Agenda for Sustainable Development. It seeks to ensure inclusive and equitable quality education for all. Digital technology have emerged as an important instrument for achieving this goal. This study highlights the importance of digital technology in education, including its applications and challenges.

Keywords: Digital Technologies, Digital Classroom, Modern Technology, Teaching-Learning.

Gen AI Models for Adaptive Content Recommendation in Online Learning platforms

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Abstract

Online learning platforms have transformed education through increased accessibility and flexibility, yet conventional recommendation algorithms cannot cope with the diverse and evolving needs of individual learners. In this paper, we propose an adaptive content recommendation framework using Generative AI models—namely Variational Autoencoders and Transformer-based architectures—to produce personalized learning sequences in real-time. Through processing big-scale real-world data from successful education platforms, our method adaptively optimizes educational content according to user interaction, learning style, and performance metrics. Experimental results demonstrate statistically significant improvements in user satisfaction and learning accomplishments compared to state of-the-art recommendation methods. Furthermore, our research addresses key implementation challenges, including data privacy, ethical data management, and explainability of algorithms. The proposed framework not only offers a replicable strategy for adaptive content recommendation but also provides useful information to instructors and developers interested in enhancing online learning experience. Index Terms— Generative AI, Adaptive Learning, Content Recommendation, Online Education, Personalized Learning

How Benchmarks Assess Mathematics: A SWOT Analysis

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Abstract

The role of technology in accelerating mathematics teaching and learning has grown significantly. Over the past decade, there has been a surge in the adoption of graphic calculators and computers. And, computers are increasingly used to access and assess online information and to create new content for math classrooms. Research indicates that strategic technology use can enhance conceptual understanding and enable access to higher-level processes without sacrificing traditional paper-pencil skills. This article investigates how emerging technologies-such as MATH, GSM8k, Frontier Math, and AI-powered chatbots are transforming mathematics education. Through a SWOT (Strengths, Weaknesses, Opportunities, Threats) analysis, it evaluates the potential and challenges of these tools in improving student engagement, personalization, and teaching efficiency in math classrooms. These advancements allow students to focus on exploration, reasoning, conjecture, and problem-solving, rather than being bogged down by rote memorization, repetitive drills, or symbolic manipulation.

Keywords and Phrases: Mathematics education, Artificial Intelligence (AI), Chatbots

A Phoneme-Based Haptic Language: Design, Implementation, and Evaluation Using a Wearable Multi-Actuator Device

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Abstract

This paper presents a novel phoneme-based haptic language system designed to support inclusive communication for individuals with hearing impairments, directly aligning with the conference theme of "Digital Transformation and Educational Technologies." Our system translates all 44 English phonemes into learnable haptic patterns using a wearable multi-actuator device, bridging the gap between speech and touch for more equitable communication in educational environments. The system systematically maps phonetic features such as articulation, voicing, and prosody to six tactile parameters—intensity, frequency, duration, rhythm, spatial location, and temporal envelope—enabling intuitive tactile feedback for spoken language. Two hardware configurations (4-actuator and 8-actuator) were developed and evaluated through controlled user studies, yielding phoneme recognition accuracies of 78.4% and 85.2% respectively. Real-time speech-to-haptic conversion with sub-100ms latency enables natural dialogue support, while the structured training protocol promotes learnability and long-term retention. This work demonstrates the viability of tactile linguistic systems in digital accessibility, opening new pathways for assistive communication technologies in educational and professional settings.

Keywords: haptic language, tactile communication, assistive technology, digital accessibility, educational technology

Active Learning

Enhancing Learning through Student-Centred Approaches

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Abstract:

In recent years, the educational landscape has undergone a significant transformation due to changes in student demographics, learning preferences, and the rapid integration of technology. Today's learners, whether at the higher secondary level or in professional education, demand more interactive, flexible, and personalized learning experiences. However, many institutions rely heavily on traditional teacher-centred methods that often fail to engage students meaningfully or support the development of critical thinking, problem-solving, and lifelong learning skills. This disconnect between outdated instructional models and modern student needs poses a significant challenge to effective learning.

This study investigates the effectiveness of a student-centred intervention implemented across two distinct learner groups: higher secondary and professional-level learners. The intervention was designed around a blended learning model that combines face-to-face instruction with digital learning tools, offering greater flexibility, interactivity, and learner autonomy. The results indicate that blended, student-centred learning environments lead to higher engagement, improved retention, and better academic performance than conventional lecture-based approaches.

Furthermore, the study highlights how the shift toward student-centred strategies better aligns with the evolving educational context where information is readily accessible, attention spans are shorter, and the demand for real-world skills is growing. It also addresses the practical challenges of adopting such approaches, including resistance to change, technological limitations, and the need for teacher training. Ultimately, this research supports the argument that student-centred, blended learning models are more relevant in today's world and more effective in fostering meaningful, lasting learning outcomes.

Keywords: Student-centred learning, educational transformation, intervention, blended learning

Novel Teaching Interventions an Approach to Ignite Cognitive Thinking: Sensitizing Brain to Process Information for Concept Building

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Abstract

Neuroscientists explain human cognition as cascading cycles of repetitive brain events. These repetitive brain events are executed in brains as proceedings of cognitive cycles, which may initiate unconsciously with a mild & temporary ignition. In order to convert this momentary ignition into a high-level conscious ignition and its registration to the brain can only take place through some non-repetitive events with an exuberating stimulus. This stimulus shall initiate a cascading cycle, which will be performed through conscious ignition leading to higher order cognition (HOC). In teaching, HOC can be explained as a state of the brain where it can have ignition of cognitive thinking leading to establishment of multiple cascading cycle operational at same time, which are consciously driven through an exuberating stimulus/signal. The multiple cascading cognition cycle shall lead to sensitization and process different subsets of available or already stored information to develop concept building in the brain. In case of students, the exuberating stimulus/ signal can be novel teaching interventions, which are far more intense in igniting HOC in comparison to traditional teaching practices. The study reveals that students, when approached through traditional teaching, perceive it as unconscious momentary ignition, which cannot convert the dispensed knowledge to run multiple cascading cycle, hence cannot achieve HOC. The study exhibit novel teaching interventions viz: Environment cognoscenti video for first year student strength 1700, peer foot printing for M.Tech students strength 30, project led teaching strength 30, were tested for sensitization and concept building in the students. The students perceived the knowledge more easily, enthusiastically with enhanced HOC. These novel interventions suggest that students can be effectively oriented to run multiple cognition cycles, which will be helpful for the student to overcome its distraction, difficulty to perceive and not to stay in an unconsciously driven mild cognitive state. In the era of fast flowing non-essential information, novel teaching intervention shall be a boon for the students to achieve ignition of higher order cognition.

Impact of Perceived Stress on Psychological Well-Being, Aggression and Emotional Regulation

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Abstract

Perceived stress has emerged as one of the most pressing concerns affecting students' psychological functioning and overall quality of life. Stress is not merely a passing discomfort but a multidimensional phenomenon with implications for emotional regulation, aggression, and psychological well-being. Although previous studies have examined these constructs individually, relatively few have investigated their combined effects within the same population, particularly in the Indian academic context. This paper examines the impact of perceived stress on emotional regulation, aggression, and psychological well-being among university students. Using standardised instruments, data were collected from 100 students (50 males, 50 females). Correlational and regression analyses demonstrated significant associations between stress and multiple psychological outcomes. Stress was negatively linked with emotional regulation and psychological well-being, but showed mixed associations across the well-being dimensions. These findings underscore the urgent need to address stress management as a central issue in student mental health. Beyond mapping correlations, the study highlights the practical and theoretical implications of understanding stress within a biopsychosocial framework, pointing toward interventions such as resilience-building, emotional regulation training, and support-based approaches as possible pathways for reducing the detrimental impact of stress.

Keywords: Psychological well-being, Emotional Regulation, Aggression, Mental health

Transforming Traditional Teaching Methods with Project-Based Learning

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Abstract

This analysis addresses the transformation from traditional teacher-centered instructional approaches toward student-centered ones by means of the implementation of Project-Based Learning (PBL). The study is based on a survey that was launched among educators at different educational levels and in diverse disciplines to explore their perceptions, experiences, and challenges in implementing PBL approaches in their teaching. According to the survey results, there seems to be a growing awareness of the weaknesses of traditional teaching approaches in building students' capacities for thinking critically, working collaboratively, and solving problems relevant to life. These findings also illustrate the desire of a vast majority of teachers to consider PBL as a good strategy to attract students' interests in learning, with the continued hindrance of time, training, and curriculum alignment. This paper compares PBL adoption according to years of teacher experience, subject area, and school support. From the perspective of the survey-based analysis, systemic changes, namely, professional development and administrative support, are required for PBL to reach its full potential in transforming educational practices. Recommendations are provided to assist educators and institutions in successful and sustainable pedagogical shift progression.

Keywords: Project-Based Learning, sustainable pedagogy, STEM education, Professional Development

Active Learning Intervention for Enhancing Problem-Solving Skills in Programming for Mixed-Discipline Students

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Abstract

Fundamental programming skills form an essential requirement for students across all fields in this contemporary data-driven world. However, the usual lecture-oriented mode of teaching generally fails to interest diverse students, especially in classes where the students exhibit varied levels of programming skill. The work on programming assignments originated from cross-channel fertilization of perspectives, which stimulated working on unique strategies for solving problems that one discipline alone cannot achieve. This paper presents an active learning method applied in introductory Programming for Problem Solving for B.E./B.Tech students of various disciplines. The method was implemented for eight weeks wherein peer mentoring was mixed with live coding demonstrations and debug team works. These methods initiated stronger student engagement with programming concepts and in turn developed students' understanding and retention skills. Feedback on the program was gathered through an anonymous survey. Evidence from surveys and focus-group discussions indicates higher engagement, improved conceptual understanding, and increased confidence in applying programming to discipline-specific problems. Although initial resistance and limited time posed challenges, the findings affirm the effectiveness of well-structured active learning in cultivating inclusive, participatory, and skills-oriented classrooms. The results argue for wider adoption of evidence-based, student-centered pedagogy across STEM curriculum. Keywords: Active learning, problem-solving, programming education, interdisciplinary teaching, flipped classroom

Beyond the ScoreBoard: Leveraging Game-Based Learning for Outcome-Based Education

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Abstract

With the advancement of Education 4.0, which is moving towards learner-centered education, Game-based Learning (GBL) has been recognized as an effective pedagogy that can increase the motivation and engagement of students for better learning. This paper investigates how the GBL strategies can be effectively aligned with the elements of Outcome-Based Education (OBE) to promote knowledge acquisition, critical thinking, collaboration, and creativity as the generic outcome that prepares students to face real-world challenges.

This paper discusses and illustrates the educational use of game-based approaches for stimulating active learning through individual and group gaming quizzes, such as Kahoot!, simulation games, role-play, peer-generated games, and educational design platforms to address particular learning outcomes in the cognitive and psychomotor aspects. Data from student comments, performance analytics, and engagement measures support the notion that game-based techniques boost not only learner participation but also long-term retention and information transfer, making the process of obtaining OBE not only productive but also enjoyable.

Implementation of Active Blended Learning Pedagogy to Enhance Student Engagement and Academic Performance: A Case Study

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Abstract

The paper explores how Active Blended Learning (ABL) methodology affects student's academic achievement through a detailed case study conducted in an undergraduate engineering course. ABL emphasizes interaction, collaboration and self-directed learning by integrating traditional face to face education with online learning activities. The study used a mixed-method approach combining qualitative feedback and quantitative evaluation of academic performance. The findings show that ABL considerably enhances understanding, retention span and problem-solving skills and creates a more engaged and supportive learning environment. The results indicate that ABL pedagogy can be a potent instrument for raising student achievement and offering insightful information to teachers who want to improve educational outcomes.

Keywords: Active learning, blended learning, collaborative learning, teaching pedagogy, learning outcomes.

Class Size and Its Influence on Student Participation During Group Work Activities

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Abstract

The effectiveness of group-based learning in large class setup and small class setup were evaluated. A topic from their curriculum was discussed and then few unsolved problems were given to the students. They had to solve the problems in group (4-5 students per group). After the activity, the feed-back of all the students were taken using a one-minute paper. The instructor played a passive role during the group discussion. After taking feedback, the role of the instructor was to resolve the unsolved issues in the group. From the feedback it was concluded that group discussion is preferred by the majority of the student provided all student participate equally and the group are formed according to the student's choice. However, from instructor point of view this task was easy to execute in a small class but in a large class it was challenging.

Keywords: group-based-learning, large-classroom, small-classroom, one-minute paper, class-size

KUAC Active Learning Framework: A Pedagogical Design for Increased Student Engagement in Large Classrooms

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Abstract

There is sufficient evidence that classrooms with fewer than 20 pupils yield significantly better academic performance than larger classrooms [1], but large classrooms are an unavoidable reality, particularly in India. AICTE reported an additional 1.4 lakh B.Tech undergraduate intake from 2023-24 to the 2024-25 academic year. [3] In such environments, lecture-based teaching strategy for large classrooms has led to cognitive passivity, nearly zero interaction of students with teachers and peers, surface-level learning and negligible engagement with the course. [4] The recent drive to restructure classrooms for improving learning has been shaped by the term “Active Learning.” [9] Bonwell and Eison’s 1991 framework on Active Learning is widely used to solve the limitations outlined above, but they themselves recognize “the difficulty of using active learning in large classrooms”. [5] We propose the KUAC Framework, to enhance student engagement in large classrooms by creating a learning environment that supplements the “traditional, instructor-focused, *Teaching by Telling* approach” [6]. The KUAC Framework was used to design and deliver a One-Credit course, titled “Personal Communication”, for first-year engineering undergraduates. In the KUAC Framework, “K” represents Knowledge being disseminated by the instructor to a large class seated in groups of six in an innovative “Horse-Shoe” seating arrangement with every group facing the stage, thus enabling group reflection during the lecture. “U” represents Understanding of concepts and theories through discussions within the “Horse-Shoe” group in the large classroom.[10] “A” represents the Activities that students participate in to apply the concepts and acquire skills. This takes place in a tutorial session conducted simultaneously for every “Horse-Shoe” group in the large classroom.[7] Finally, “C” represents Communication of the student’s learning presented by the students to their “Horse-Shoe” group. The findings of our study revealed that the KUAC Framework (a) considerably increased peer-peer and faculty-student engagement, (b) brought about a deeper engagement of concepts and their supervised application in the classroom, and (c) optimized student learning as it was dialogically communicated to both their peers in the small group and faculty in the classroom.

Keywords: Active Learning, Classroom Pedagogy, Large Classroom

Revamping the Tertiary Education System: Challenges and Opportunities in the Student-Centric Interactive Teaching and Learning Processes

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Abstract

Currently, the fast-changing socio-economic and geopolitical landscape is profoundly impacting the well-being of not only billions of us but also other life forms of this planet every day. A plethora of anthropogenic activities like deforestation, pollution, industrialization along with the geopolitical conflicts are aggravating our surroundings, societies, and value systems. In such an ever-changing ecosystem, we cannot live today the way we lived yesterday. The need of the hour is to identify the root causes of challenges for possible remedial measures. In recent years, poverty, social unrest and employability of young generation, in particular, remain major challenges in almost every society. Countries all over the world are now putting emphasis on restructuring the education system on multiple aspects at all levels, particularly tertiary (higher) education considering its immense potential in human capital development and innovation. The core principle of the Sustainable Development Goal 4 (SDG 4) is to “ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.” The National Education Policy 2020 (NEP 2020) and Education for Sustainable Development (ESD) are aligned with SDG 4 to respond various challenges with stipulated local and global goals, guidelines, and solutions. The need of the hour is to revamp our education systems through strategically and synergistically tailoring academic policies and student-centred learning approaches. However, passivity, reluctance towards learning new skills/ideas, lack of inquisitiveness, lack of active engagement and motivation are still some common challenges in educational settings. A sensible, caring, compassionate, knowledgeable, open-minded and professionally-groomed teacher with empathy can create a conducive academic environment to overcome such difficulties through multiple pedagogical approaches. In this connection, the Industry 4.0 technologies are relevant and useful in shaping and implementation of Education 4.0.

Keywords: Major global challenges; Revamping tertiary education, Sustainable Development Goal 4; Education for Sustainable Development; Student-centred learning; Pedagogical approaches

Role of Technical and Non-Technical Societies in Shaping the Future of Engineers

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Abstract

In the present academic prospect, the non-academic involvement is becoming progressively critical while influencing the wide-ranging growth of future engineers. Non-technical and technical cultures of societies, along with various clubs of professional bodies, are pivotal platforms that are nurturing essential skills beyond the classroom. The main emphasis of this research work is the insightful impression of technical societies such as IET, IEI, ASME, IEEE, SAE, and ISTE in training engineering students with problem-solving capabilities, innovative familiarity, and applied acquaintance through seminars, workshops, competitions, industrial collaborations, and hackathons. The culture of Non-technical and technical societies acts as a bridge between academic models and real-world presentations. With the innovation being fostered, along with interdisciplinary learning and timely research exposure, the cognitive and incognito vulnerabilities among students faced by students while switching from academic to social or professional life are reduced. Involvement of students in these enhances procedural expertise, leadership, communication, and management skills, which play a vital role in the engineering profession. In addition, non-technical societies such as social forums, literary clubs, debate societies, and entrepreneurship cells play a balancing role in the individual growth of engineers by humanising emotional intelligence, communication skills, and resourceful thinking. The above attributes are gradually appreciated in a global and team-oriented environment. Furthermore, societies of national and international repute craft opportunities for global mentorship, networking, and exposure to global inclinations and morals. Students are encouraged to understand the community and provide engineering-based solutions, take part in workshops, seminars, conferences, and publish papers. Jointly, these clubs and societies create a practical ecosystem that aligns the academic learning with industrial needs and societal challenges. Also, to foster a mindset of continuous learning with eagerness to handle social responsibility. While dynamically appealing to both non-technical and technical societies, engineering students appear not just as technically equipped professionals, but also as adaptive, accountable, and progressive individuals. Hereafter, the assimilation and inspiration of such societies within academic institutions is vital in determining the future of proficient and accomplished engineers.

Keywords: Technical Societies, Non-Technical Societies, Engineering Education, Skill Development, Professional Growth

The Role of Academic Group Projects in Shaping Professional Teamwork Competence: A Gender Perspective

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Abstract

Group work has become a crucial way of learning in the modern education system. These new teaching methods equip students with essential skills such as problem-solving, conflict resolution, accountability, teamwork, critical thinking, and collaborative learning. These collaborative academic experiences also foster abilities like navigating network diversity and functioning effectively in heterogeneous teams. Beyond the classroom, these skills are crucial for the professional growth of all students. This study explores the long-term impact of academic group projects on professional teamwork. This study is focused on gender-based differences among students who previously engaged in group work during their academic years. The target demographic includes individuals between the ages group 21 to 28. The research aims to assess how group projects contribute to the development of individuals and professional traits and how these impacts vary between male and female participants. It particularly explores the qualities that group projects help develop, such as multitasking, creative thinking, and many more, which may be influenced by the group of professionals and how such projects may help both genders in strengthening their inherent abilities. Additionally, the study offers insights into improving teaching strategies to better support students' development of real-world competencies through group work.

Keywords: Academic Group Projects, Professional readiness, Team Work, Students development, heterogeneous teams, Collaborative learning.

Impacts of Class Attendance and Reviewing Evaluated Papers on Academic Performance in a 1st Year Engineering Chemistry Course

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Abstract

Previous research supports the positive relationship between class attendance and academic performance, albeit a larger number of studies sampled core courses. However, the relationship between these two variables for the non-core courses is not much studied. Further, the associations between students' habit of checking evaluated answer papers, class attendance, and academic performance are not well known. To bridge these gaps, this correlation study investigated the nature of associations between the three-abovementioned variables for a Chemistry course taught in the engineering discipline in a class of 225 students. The results of Spearman's rank correlation showed a significant positive association between the total marks acquired in the mid-semester test (MST), sessional and practical (MSP), and reviewing of the evaluated MST paper ($\rho = 0.28$). Significant positive associations were also found between total attendance (TA) and both reviewing of the evaluated MST ($\rho = 0.3$) and end-semester test (EST) ($\rho = 0.14$) papers. Further, total marks (TM) showed a significant positive relationship with TA ($\rho = 0.37$). Interestingly, our results also revealed that students who turned up to check their evaluated MST paper scored significantly higher in the MST than those who did not turn up ($U = 4153$, $p < 0.0001$). This effect, however, disappeared for the EST, which is discussed in the paper. A linear regression analysis showed TA to be a significant predictor of TM, with 12% of the variability in TM explained by TA, and a 1 standard deviation increase in TA led to a 0.34 standard deviation increase in TM. Altogether, our results show that students with higher total attendance had the habit of reviewing their evaluated answer papers and acquired higher total marks. The impacts of class attendance and reviewing of the evaluated answer papers on the scholastic performance are discussed, which drive the academic trajectory of students.

Keywords: Class attendance; Paper reviewing; Academic performance; Student performance; Non-core courses, Mid-semester test, End-semester test

‘Brainaculum’ – A Novel Neuropedagogical Paradigm to Match Curriculum and Pedagogy with Human Cognitive and Brain Functions - Model and Case Study

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Abstract

The current educational models often have limitations in offering conceptual clarity and adequate practical training to the students. Further, a healthy body of research has demonstrated increased academic stress in students, which is associated with compromised motivation and creativity, low academic success, and severe psychological conditions. Importantly, the designs of curricula and pedagogical techniques in these educational models are also in serious inconsonance with the cognitive and brain functions of humans. To mitigate these limitations, we seek help from cognitive psychology and neuroscience to transform engineering and nonengineering education. Bridging the fields of neuroeducation, instructional design, and cognitive psychology, we propose a model, called Brainaculum, which is a human cognitive and brain function-centric design of curriculum and pedagogy. Aligning education with neuro-cognitive principles, Brainaculum maps curriculum and pedagogy on some of the pivotal features of human cognition, related to scholastic activities; learning is purposive, cognitive mutualism, repeated activations of memory networks in the brain, working memory limits, building of ‘schema’ during knowledge acquisition, spacing repetitions of highly-structured academic information, emotional-cognitive synchronization, training programs to improve cognitive abilities, etc. The model, Brainaculum, has the goals to significantly enhance abstract and experiential learning, reduce the overall academic pressure, and reinforce positive mental health in the students, which will inevitably contribute to their academic motivation, learning and memory, creative thinking, self-esteem, self-perceived happiness, and management of stress. Moreover, Brainaculum promotes the training of teachers with the tools of educational psychology and neuroscience so that teaching styles can match the preferred learning styles of students, teachers can guide the students to find out appropriate study strategies for various courses, and foster students’ capacity to become ‘abstraction’ learners. This way, Brainaculum intends to contribute to the various dimensions of the ‘Neuropedagogical’ paradigm.

Keywords: Brain; Brainaculum; Neuro-pedagogical paradigm; Neuroeducation; Curriculum; Pedagogy; Academic Stress; Education Psychology; Instructional design; Emotional-cognitive synchronization; Experiential learning; Learning Style Inventory; Abstraction Learners

Bridging the Gap between Curriculum and Competence: The Promise of Outcome-Based Education

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Abstract

The global economy's rising dynamism and skill requirements has created a major issue because academic programs do not match industrial requirements. The traditional educational approach which focuses on content delivery and teacher instruction does not provide students with the necessary practical skills and critical thinking abilities needed for professional success. This paper evaluates the strategic role of Outcome-Based Education (OBE) in resolving the curriculum-competency mismatch between education and industry. The educational process under OBE focuses on specific learning outcomes while matching teaching methods to these objectives and requiring students to demonstrate their achievements. The research starts by studying the fundamental principles and philosophical base of OBE which includes backward curriculum design and learner autonomy and continuous improvement. The paper demonstrates how institutions should develop and connect their curriculum mapping with program educational objectives (PEOs) and program outcomes (POs) and course outcomes (COs). The paper examines effective implementation strategies in institutions while discussing obstacles including faculty opposition and infrastructure constraints and insufficient policy backing and presents possible solutions.

The paper dedicates attention to how accreditation bodies including NBA and ABET establish OBE standards while emphasizing the necessity of faculty development programs and feedback systems and data assessment tools. The paper outlines a step-by-step approach to implement OBE in higher education institutions which will achieve academic excellence and develop employable graduates who can compete globally. The research supports a complete educational system change by promoting outcome demonstration instead of content memorization. The study provides practical implementation strategies and strategic guidelines for educational institutions to establish Outcome-Based Education in a sustainable and effective way.

Keywords: Outcome-Based Education, Curriculum Design, Competency Development, Employability Skills, Academic-Industry Gap

A Study on Group Learning: Implications to the Think-Pair-Share Strategy

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Abstract

Group learning is an effective approach and valuable tool in promoting active learning. This study explores group learning and its implications using the think-pair-share strategy. In this study, we aimed to understand how the think-pair-share strategy supports student-centred learning, enhances student engagement and responsibility for their own learning, and creates an inclusive environment that allows learners of different abilities to participate actively. Current work uses this intervention in a second-year chemical engineering tutorial class. In the current work, students were separated into groups of six, with four students in each group. We examined three cases. The first scenario required students to tackle tutorial questions without group discussion. The second scenario required students to address tutorial problems using think-pair-share in their own and other groups. The instructor handled tutorial problems monologue-style in the third case. A survey was conducted, and students were asked to rate their learning in the above three cases on a scale of 4 to 1, with 4 being the highest degree and 1 being the poorest. Based on the survey data, a weighted average score is computed. The weighted average score for the first case is 2.27, for the second case 3.29, and for the third case 2.30. The higher value of the weighted average score of the second case indicates that the think-pair-share learning strategy proved comparatively better. The findings suggest that think-pair-share increases student confidence, promotes a more profound understanding of content, improves peer interaction, and fosters a collaborative learning culture.

Keywords: Group learning, active learning, think-pair-share, monologue, peer interaction.

Teaching Together, Learning Together: Mentorship-Driven Co-Teaching for Academic Innovation

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Abstract

In the academic landscape, co-teaching supported by mentoring has become a vital approach for transformative education. This pedagogical model fosters collaborative learning, encourages reflective practice. This study investigates the impact of mentoring in a co-teaching environment through a structured program involving 30 faculty cohorts and one experienced mentor. The focus was on exploring ethical concerns, policies, and implications of artificial intelligence (AI) in education—a multidisciplinary topic requiring diverse perspectives and critical inquiry.

The co-teaching sessions enabled peer-to-peer knowledge exchange, the development of innovative teaching strategies, and enhanced faculty engagement. Cohorts participated in various joint activities including designing and delivering microteaching sessions, co-developing assessment strategies and creating multiple mini-assignments for integration of these experiences into undergraduate curricula. Participants reported a marked improvement in their confidence to address complex ethical topics, adoption of inclusive pedagogical techniques, and a deeper understanding of AI's socio-technical dimensions. A key finding was that equal status among co-teachers and the presence of a supportive mentor created a psychologically safe space for experimentation and mutual learning. It was also found that Effective co-teaching required high levels of self-discipline, mutual respect, accountability, and professional alignment.

The study concludes that mentoring in co-teaching can act as a powerful enabler of transformative education. Institutional support in the form of policy, recognition, and training is essential for the scalability and sustainability of such initiatives. The findings offer a roadmap for embedding co-teaching frameworks in faculty development programs.

Strategic Implementation of Outcome-Based Education in Higher Education Institutions

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Abstract

Recently, there has been a dynamic shift from traditional education system (TES) to outcome-based education (OBE). OBE is a student-centered method that focuses on the specific outcomes that students are expected to attain at the end of each instructional time. OBE focuses on what learners learn and display in terms of knowledge, skills, and attitudes rather than typical educational inputs such as class time or subject covered. The measurable learning outcomes form the basis for developing and aligning all areas of the educational process, such as curriculum development, teaching tactics, resource selection, and assessment methodologies. In practice, the teaching-learning process (TLP) can be evaluated using both qualitative and quantitative measures. At each stage of OBE, the quantification of outcomes ensures that every component of instruction is consciously designed to assist students in achieving the targeted results. Also, mission and vision are a self-evaluation process for the institution. Moreover, this work provides a comprehensive evaluation and process for applying OBE in higher education institutions.

Keywords: Outcome-Based Education, Blooms taxonomy, Course Outcomes, Program's outcomes, student-centered approach, teaching-learning process.

Enhancing Student Engagement through Peer Feedback in a Large Classroom Setting

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Abstract

Large classroom engagement has been a great challenge in the teaching all over the globe. Providing timely and effective feedback is a common challenge in large classroom settings. It often carries defiance student's community. Traditional feedback methods often result in delays, limiting students' ability to improve their work before final assessments. Peer feedback can be an effective strategy to enhance learning, as it promotes critical thinking, self-assessment, and collaborative learning. This paper is trying to address this challenge by incorporating structured peer feedback sessions. Students were required to review and provide real constructive feedback on their peers' assignments using a rubric-based approach. The goal was to create a collaborative learning environment where students could learn from one another while improving their ability to understand and refine their own learnings.

Transforming Programming Pedagogy through Structured Active Learning Strategies in Programming Course

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Abstract

Conventional approaches to teaching programming courses, such as Python, C++, etc., utilize passive delivery of instructions and disconnected lab sessions, which impairs student engagement and suppresses collaborative learning. To resolve these issues, an active learning model was adopted for the cultivation of an interactive and student-oriented environment. Different strategies were incorporated, namely poll quizzes, Pedagogical Shift in Student Involvement Study (PSIS), evaluations through quizzes, and collaborative learning using Student Teams Achievement Divisions (STAD) to promote peer-learning and motivation among students. Also, rubrics were designed and shared with students prior evaluations to encourage reflective practice, ensuring transparent and fair assessment. Further, a few assignments were mapped according to the ARCS learning model to accomplish student confidence, attain the attention and overall satisfaction. The implemented approach resulted in significant improvements in the conceptual understanding of the topics, leading to the adoption of ethical coding and improved student participation. The current study presents an adaptable learning model by implementing active learning activities in the coding courses, promoting interactive learning with better outcomes.

Optimizing Students' Engagement by Class Size: Review of Large vs. Small Classrooms under Education 4.0.

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Abstract

While we are in an ever-increasing educational digitalization and pedagogical innovation age, students cannot simply be spectators of their education if they are looking for academic success and cognitive development. This study is designed to identify the effect of class size on students' participation. In particular, the research contrasts big classes that rely on lectures with smaller ones that are interactive. The research is rooted in Attention Theory and Active Learning Principles and employs an AB quasi-experimental method. It also contains 100 undergraduates who have experienced Lecture A (mega) and B (small) rooms.

Upon completion, the degree of users' attention, motivation and cognitive engagement will be measured with standard engagement instruments. The relationship between class size and engagement variation will be analyzed quantitatively to determine significant patterns. The findings should provide empirical evidence for improving forms of classes by meeting the E4.0 and the OBE (outcome based education) framework. This study demonstrates strategic curriculum redesign by integrating DLT (digital learning technology) to enhance pedagogical and learning activities in various educational contexts.

Keywords: Student engagement, class size, attention, active learning, education

Tutoring and Mentoring Fostering Constructive Paradigm Shift in Higher Education

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Abstract

Tutoring and mentoring has evolved over the decades as a powerful tool in the constructive progression of higher education. This paper investigates how mentorship and tutoring might help start a positive paradigm change in higher education. Traditional models of education frequently fall short when institutions are under more and more pressure to adjust to changing student demands, a variety of learning styles, varied industry demands, and the rapid advancement of technology. Based on individualized instruction, peer assistance, and faculty involvement, the study looks at how structured tutoring and mentoring programs improve student learning results, boost retention, promote equity of power and cultural and regional tolerance, and foster inclusive academic cultures. Tutoring and mentoring are also transformative forces in dealing with the students' anxiety and stress issues. The study makes the case supported by case studies and empirical data that these support networks empower students, encourage critical thinking, and build academic resilience in addition to enhancing formal curriculum. In the end, it can be safely stated that mentoring and tutoring are portrayed as stimulants for a more student-centered, flexible, and comprehensive strategy for reforming higher education on one hand, and providing compassionate and empathetic hand holding environment for the wholistic growth of its stakeholders on the other.

Enhancing Collaborative Problem Solving in Engineering Education through Think-Pair-Share: Implementation and Insights

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Abstract

This study explores the effectiveness of the Think-Pair-Share (TPS) technique as an active learning strategy to enhance collaborative problem solving, student engagement, and critical thinking across diverse educational contexts. Initially implemented in core engineering courses (e.g., Electrical & Electronics Engineering, Physics), TPS was structured around three phases: individual thinking, peer collaboration, and group discussion. Activities ranged from analytical problem-solving using simulation tools to design-focused project tasks. The intervention was evaluated using a mixed-methods approach: pre and post-performance comparisons, tutorial mark analyses, and reflective feedback (e.g., “Minute Papers”). Results demonstrated significant improvements in conceptual understanding, academic performance, and participation, particularly among students who were previously disengaged. Faculty discussions and student reflections indicated increased confidence, motivation, and inclusivity in classroom dynamics. Encouraged by its success, TPS was scaled to interdisciplinary and project-based courses (e.g., Engineering Design Projects I & II), validating its adaptability beyond technical content. Although challenges in large lectures (e.g., coordination and noise) were noted, the positive spillover from small-group interaction was evident. Rooted in the New Directions Programme and supported by pedagogical literature, this study positions TPS as a scalable, evidence-based model for transforming traditional classrooms into active, student-centered learning environments across STEM, basic sciences, and management education.

Keywords: Active learning, collaborative learning, educational technology, teaching innovation, Think-Pair-Share (TPS).

Academia-Industry Connect

‘Valley of Death’: A Case for Bridging the Gap for Trans-Disciplinary Research in India

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Abstract

The debate around academia-industry relationship is not new but it has intensified in the last half a century. It gained momentum after WW-II when the policy makers around the world raised their expectations from the sponsored research. The governments started seeing universities as not only centres for learning but also as 'innovation hubs' and 'engines of economic growth'. The idea was implemented in the US through the Bayh-Dole Act, 1980 which allowed the universities to commercialise the government funded research. With the ever rising demand from the industry for the market ready skilled manpower the subsequent policies of the government, from the 'Programme of Action, 1992 to Skill India Mission, 2015 and NEP 2020 have all emphasised the need for stronger ties between the industry and the academia. The Department of Science and Technology, took the initiative in 2014-15 by establishing Technical Research Centres to bridge the gap between the Scientific Research and its Real World application. Since the debate has now moved from 'should we collaborate to how can we collaborate?' especially in terms of encouraging Scientific Research to achieve Sustainable Development Goals. Government is expected to play the pivotal role in bridging this 'Valley of Death' by increasing the investments into Applied Research. Therefore, the endeavour of this paper is to highlight the need for Trans-disciplinary Research facilities to provide a platform for research into interconnected social, environmental and economic issues. It will not only strengthen the industry - academia relationship but also facilitate confluence with effective policy-making.

Keywords: Scientific Research, Applied Research, Valley of Death, Trans-Disciplinary Research, Innovation

AI in Mathematics Education: A Critical Review of Its Role, Impact, and Pedagogical Implications in Higher Education

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Abstract

The integration of large language models like ChatGPT into mathematics education marks a paradigm shift in how learners engage with abstract reasoning, problem-solving, and proof-writing. This review paper critically examines the role of ChatGPT in supporting mathematical learning in higher education settings. By analyzing use cases such as solving problems, generating explanations, assisting with mathematical writing, and exploring concepts interactively, we evaluate both the pedagogical benefits and the limitations of this technology. Drawing from recent studies, classroom observations, and theoretical insights, the paper highlights how ChatGPT can enhance cognitive engagement, foster inquiry-based learning, and support differentiated instruction. Challenges around accuracy, student dependency, and epistemological concerns are also addressed. The paper concludes with recommendations for responsible integration of AI in the university-level mathematics classroom.

Keywords: ChatGPT in Education, Artificial Intelligence in Mathematics Learning, Large Language Models, AI-Assisted Problem Solving, AI and Mathematical Reasoning.

From Classroom to Cubicle: Decoding the Internship Experience as a Career Catalyst

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Abstract

As the global job market becomes increasingly competitive, bridging the gap between academic training and professional preparedness is more critical than ever. This research investigates the transformative role of internships—particularly off-campus, year-wise experiences—in shaping students' career trajectories and professional identities. Drawing on a longitudinal, mixed-methods study involving multiple participants across multiple academic years, the project combines ethnographic workplace observations, digital portfolio analysis, in-depth interviews with industry mentors, and year-wise career tracking to capture a holistic picture of internship impacts.

Findings reveal that internships are not merely supplementary to academic curricula but serve as essential platforms for the development of professional competencies, adaptability, and industry-specific skills. Second-year interns, embarking on their first real-world experiences, show marked improvements in workplace confidence and foundational skills, while third-year students demonstrate deeper integration of theoretical knowledge into practical applications. Fourth-year participants, engaging in capstone internships, exhibit significant advantages in career placement and are often better positioned for early professional growth compared to their non-intern peers. The study identifies mentor engagement, authentic project responsibilities, and exposure to dynamic work environments as key factors that drive professional maturation during the internship period.

Additionally, the research highlights how off-campus internships facilitate vital professional networking opportunities and foster a sense of industry belonging, which traditional classroom settings often lack. The longitudinal design of this study offers unique insights into the cumulative effects of internships, revealing that their value increases as students' progress through higher education, provided these experiences are well-structured and mentored. Ultimately, this research advocates for the institutionalization of off-campus, year-wise internships as integral components of higher education, offering actionable recommendations for universities, employers, and policymakers to maximize the developmental potential of these experiences for the holistic success of future professionals.

Keywords: Internship Programs, Career Development, Professional Identity, Longitudinal Study, Off-Campus Learning

Project-Based Learning and Employability: A Cognitive Constructivism and Transformative Learning Perspective from a Developing Country

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Abstract

Based on a project-integrated intervention in the Banking & Financial Services module at a management institute, this study examines how project-based learning (PBL) influences the employability of management graduates. This study was conducted over a 16-week semester, and the intervention was adopted into the curriculum by assigning real-world banking problems to 12 student groups enrolled in the elective course during the MBA programme. A standardised questionnaire was administered through Google Forms to the student groups to understand the impact of this PBL on enhancing their employability during the campus placement at the end of the programme.

By employing a qualitative thematic analysis, this study empirically tested the role of active and student-centred learning, problem-solving approach, continuous peer and instructor feedback for bridging the gap between theoretical concepts and their application in industry, and employability skills in the knowledge-intensive Finance domain of the MBA program. Building upon cognitive constructivist learning and transformative learning theories, this paper is a maiden attempt to establish the application of guidelines of New Education Policy (NEP 2020) formulated by the Government of India in the setting of management education at the postgraduate level. The findings highlight how PBL enhances employability through the mediating variables of constructivist and transformative learning. It not only adds to andragogy literature in management education by proposing a conceptual framework but also make pertinent recommendations concerning the potential of evidence-based active learning for shaping future-ready management graduates as a practical implication.

Keywords: Project-based Learning, Constructivist Learning Theory, Transformative Learning Theory, Employability



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